MILITARY SPECIFICATION

# INSTALLATION AND ACCEPTANCE TESTING OF RECEIVER, RADIO R-1963/ARN

This specification is approved for use by US Army Aviation R&D Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scone. This specification provides installation, equipment performance, engineering and production approval instructions and requirements for installation and testing of Receiver, Radio R-1963/ARN and associated equipment in prototype and production model aircraft.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Aviation R&D Command, ATTN: DRDAV-EKS, St. Louis, MO 63166 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5826

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SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-C-172 | Case; Bases; Mounting; and Mounts, Vibration |
| MIL-B-5087 | Bonding, Electrical, and Lightning Protection, for Aerospace Systems |
| MIL-W-5088 | Wiring, Aircraft, Selection and Installation of |
| MIL-E-5400 | Electronic Equipment, Airborne, General Specification for |
| MIL-E-6051 | Electromagnetic Compatibility Requirements, Systems |
| MIL-C-6781 | Control Panel: Aircraft Equipment, Packer Console Mounted |
| MIL-I-8700 | Installation and Test of Electronic Equipment in Aircraft |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-C-45662 | Calibration System Requirements |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-461 | Electromagnetic Interference Characteristics, Requirements for |
| MIL-STD-463 | Definitions and System of Units, Electromagnetic Interference Technology. |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-877 | Antenna Subsystems, Airborne, Criteria for Design and Location of |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity eras directed by the contracting officer.)

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# 3. REQUIREMENTS

## 3.1 General

3.1.1 Acceptance. Any installation of Receiver, Radio R-1963/ARN which is in accordance with MIL-I-8700 and the requirements of this specification shall be considered acceptable. Such acceptance, however, does not constitute safety-of-flight approval.

3.1.2 Responsibility for miscellaneous components. The contractor shall furnish and install all required plugs, cables, brackets, connectors, junction boxes, terminal boards, circuit breakers, switches, relays, wiring, shafts, lamps and miscellaneous items to complete satisfactory installation and operation of the Receiver, Radio R-1963/ARN. All contractor furnished electronic equipment shall be in accordance with the applicable electromagnetic interference and compatibility requirements in MIL-STD-461 and the requirements in MIL-E-5400 for Class I-A equipment designed for operation in both aircraft and helicopters.

3.1.3 Quality assurance. The contractor shall perform all examinations and tests specified in section 4 to insure compliance with the applicable requirements of section 3.

3.1.4 Radio frequency assignment. The contractor shall request from the procuring activity a radio frequency assignment for which he has no assignment and is otherwise necessary to verify that the Receiver, Radio R-1963/ARN performance is in accordance with the requirements specified herein. Receipt of radio frequency assignments from the procuring activity does not relieve the contractor of any responsibility for obtaining a Federal Communications license to operate equipment radiating electromagnetic energy.

3.1.5 Reports. The contractor shall prepare and submit to the procuring activity all reports and other supplementary data as specified by the contracting officer.

3.1.6 Check-off list (See Figure 1). The contractor shall record the results of the applicable examinations and tests specified in section 4 on an appropriate check-off list similar to the examples shown in Figures l-A, 1-B and 1-C. The contractor shall Prepare a check-off list for each Receiver, Radio R-1963/ARN installed in an aircraft and maintain a file of these records for reference, as necessary, by the procuring activity. (See Appendix)

3.1.7 Equipment non-compliance. If the equipment fails to comply with the requirements of 3.3, 3.4 or 3.5, record the failure data on DA Form 2407. In accordance with the provisions of the Reliability Improvement Warranty (RIW) Program, the equipment shall be returned for repair to Bendix Avionics Division, 2100 N. W. 62nd Street, P. O. Box 9414, Fort Lauderdale, Florida 33310. Do not break or tamper with the warranty seals as it may negate the equipment contractor repair obligation.



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3.1.8 Procuring activity approval. In all cases where approval by the procuring activity is specified, the contractor shall be in receipt of said approval Prior to conducting any tests, inspections or otherwise utilizing the proposed equipment or procedures.

3.1.9 Test equipment calibration. The contractor shall prepare a statement which identities the test equipment utilized in the performance of the tests specified in section 4. The statement shall include certification of the test equipment calibration in accordance with MIL-C-45662 and be forwarded to the procuring activity as supplementary data to the applicable test reports specified herein.

3.1.10 Electronic sub-system certification. The contractor shall prepare a statement which certifies that all aircraft installed electronic sub-systems which are utilized in the performance of the tests specified in section 4 are in accordance with the applicable specifications. The statement shall be forwarded to the procuring activity as supplementary data to the applicable test reports specified herein. The procuring activity is responsible for final approval and certification.

3.1.11 Electromagnetic compatibility. Interference control and test plans shall be prepared by the contractor in accordance with MIL-E-6051 and MIL-STD-463 and submitted to the procuring activity for approval. After receipt of approval, the contractor shall conduct the tests specified in 4.3 to demonstrate compliance with MIL-E-6051 and to insure interference free operation of the Receiver, Radio R-1963/ARN after installation in an aircraft. A contractor prepared electromagnetic compatibility test report shall be forwarded to the procuring activity as directed by the contracting officer. EMI/EMC characteristics of the Receiver, Radio R-1963/ARN which verify compliance with the applicable requirements of MIL-STD-461 will be provided by the procuring activity as required.

3.1.12 Equipment to be tested. The equipment listed in Table I shall meet the requirements specified herein. Whenever the model or detail specification for a particular aircraft provides a list not in agreement with Table I, the list in the aircraft specification shall govern.

TABLE I

ELECTRONIC EQUIPMENT TO BE TESTED
for Receiver, Radio R-1963/ARN

| DESIGNATION | NOMENCLATURE | NO. OF UNITS REQUIRED | WT. PER UNIT (POUNDS) | UNIT HEAT DISSIPATION (MAX) |
|---|---|---|---|---|
| Receiver, Radio | R-1963/ARN | 1 | 4.0 | 7.5 Watts |
| Mounting Base, Electrical Equipment | MT-4835/ARN | 1 | 0.9 | |



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3.1.13 Additional equipment required. In addition to the equipment listed in Table I, the following equipment, listed in Table II, is required for a complete installation of the Receiver, Radio R-1963/ARN. Whenever the aircraft model or detail specification for a particular aircraft provides an equipment not in agreement with Table II, the list in the aircraft specification shall govern.

TABLE II

ELECTRONIC EQUIPMENT ASSEMBLY TABLE
for RECEIVER, RADIO R-1963/ARN

| EQUIPMENT DESIGNATION | PART NO. OR NOMENCLATURE | NO. OF UNITS REQUIRED | WT. PER UNIT (POUNDS) | REMARKS |
|---|---|---|---|---|
| Switch, HI-LO Sensitivity | --- | 1 | --- | SPST switch that provides HI/LO sensitivity selection. |
| Antenna, Glideslope | AT-326( )/ARN | 1 | 0.5 | Capable of receiving glideslope signals over 329.15 MHz through 335.00 MHz frequency range. Impedance of 52 ohms. |
| Antenna, Marker | AT-640( )/ARN | 1 | 0.8 | Capable of receiving 75 MHz marker signals. Impedance of 52 ohms. |
| Control, Radio Set | C-6873B/ARN-82 | 1 | 1.5 | Provides glideslope 2-out-5 frequency selection and marker beacon test signal. |
| Indicator, Course | ID-1347C/ARN-82 | l to 3 | 3.0 | Glideslope deviation load 150-0-150 microampere, 1000 ohm meter movement. Glideslope flag load 250 microampere, 1000 ohm meter movement. |
| Lamp(s), Marker | MS-25041 | l or 3 | --- | Comprised of either a single green lamp system or a three lamp system consisting of a blue (400 Hz) lamp, an amber (1300 Hz) lamp, and a white (3000 Hz) lamp capable of operating at approximately 28 Vdc. |



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TABLE II CONTINUED

ELECTRONIC EQUIPMENT ASSEMBLY TABLE
for RECEIVER, RADIO R-1963/ARN

| EQUIPMENT DESIGNATION | PART NO. OR NOMENCLATURE | NO. OF UNITS REQUIRED | WT. PER UNIT (POUNDS) | REMARKS |
|---|---|---|---|---|
| Control, ON-OFF/ Volume | --- | 1 | --- | Combination SPST switch for GS/MB RECEIVER ON-OFF function and 150 ohm potentiometer for MB audio volume control. |

3.1.14 Test equipment required. The following test equipment or equipment of equal or superior characteristics shall be used in the performance of the tests specified herein.

TABLE III

TEST EQUIPMENT AND ACCESSORIES

| NAME | PURPOSE | PREFERRED TYPE | |
|---|---|---|---|
| | | TYPE DESIGNATION | CHARACTERISTICS |
| Rf Signal Generator | Generate rf carrier for glideslope tests. Generate modulated carrier for marker beacon tests. | Hewlett-Packard, Model 8640B, with options 001 and 004 | Frequencies of 75 MHz, and 329.15 to 335.00 MHz. Output level variable from 5 to 50,000 microvolt. Output impedance of 50 ohms. Carrier modulation of at least 95% with 400 to 3000 Hz internally generated tone or with externally generated tone. |
| ILS Audio Generator | Generate 90 Hz and 150 Hz tones for modulating the rf signal generator to produce the standard LOC and glideslope signals. | Bendix Avonics Division, Model RIG-32A | Frequencies of 90 Hz and 150 Hz with phase accuracy of ±2% at simultaneous zero crossing in equal direction of 90 Hz and 150 Hz signals. Frequency accuracy of ±. 005%. Output impedance of less than 0.1 ohm. |



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TABLE III CONTINUED

TEST EQUIPMENT AND ACCESSORIES

| NAME | PURPOSE | PREFERRED TYPE | |
|---|---|---|---|
| | | TYPE DESIGNATION | CHARACTERISTICS |
| 6 dB Attenuator Pad | Match impedance and attenuate output of HP-8640B rf signal generator. | Boonton Radio, Model 505B | Attenuation of 6 dB. Impedante of 50 ohms. |
| Radio Test Set | Provide GS/MB flag meters and indicators plus interconnection of equipment for test. | AN/ARM-92B | Contains: 150-0-150 microampere, 1000 ohm deviation meter; 500 microampere, 1000 ohm flag meter; ILS deviation indicator; control panel for receiver, and a three lamp marker beacon lamp system. |
| Interface Cable | Provide interface between AN/ARM-92B test set connectors and R-1963/ARN receiver connectors. | CX-13035/AR | --- |
| Regulated Dc Power supply | Supply dc power to AN/ARM-92B test set. | Hewlett-Packard Model 6266B | Dc output of +27. 5V at 2 amperes. Regulation of ±1%. Ripple of less than 1 millivolt. |
| ILS/MB Ramp Test Set with Transmitting Antenna | Generate glideslope and marker beacon signals for checking system operation on ramp. | Collins Radio Model 972Q-4 P/N 622-2211-002 | Glideslope: Frequency of 334.7 MHz and 334.55 MHz ±. 005%. Modulation at zero DDM of 40% at 90 Hz and 40% at 150 Hz. DDM of 0.175 for up deflection and down deflection.<br>Marker Beacon: Frequency of 75 MHz ±. 005%. Modulation 95% at 400 Hz, 1300 Hz, and 3000 Hz. |



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TABLE III CONTINUED

TEST EQUIPMENT AND ACCESSORIES

| NAME | PURPOSE | PREFERRED TYPE | |
|---|---|---|---|
| | | TYPE DESIGNATION | CHARACTERISTICS |
| Audio Output Meter | Measure audio output power of marker receiver | TS-585 ( )/U | Power measurement range to 250 milliwatts. Load impedance of 150 ohms. Accuracy of ±3%. |
| Vacuum Tube Voltmeter | Measure dc input voltage | Hewlett-Packard, Model 410C | Voltage measurement range to +30 Vdc. Input impedance of 100 megohms. Accuracy of ±2%. |
| Headset | Check audio output of marker receiver | --- | Any high performance headset. |

3.2 Installation (See 4.4). All Government furnished aircraft equipment and contractor furnished equipment shall be installed in accordance with MIL-I-8700 and as specified herein.

3.2.1 Electrical wiring. (See Figures 2 and 3). All electrical wiring required for installation of Receiver, Radio R-1963/ARN in an aircraft shall be in accordance with MIL-W-5088 and as specified herein.

3.2.2 Electrical bonding. Electrical bonding shall be in accordance with MIL-B-5087 and as specified herein. Refinishing of bonded surfaces shall be in accordance with the applicablc requirements of MIL-F-14072.

3.2.3 Antennas. Installation, design and location of airborne antennas shall be in accordance with MIL-STD-877 and as specified herein.

3.2.4 Control panels. All contractor furnished rack or console mounted control panels shall be in accordance with MIL-C-6781 and as specified herein.

3.2.5 Mountings. All contractor furnished cases, mounting bases and vibration mounts shall be in accordance with MIL-C-172 except that the transmissibility in each of the three mutually perpendicular principal axes of the mounting base shall not

3.2.6 Precautions.

a. Do not disconnect or connect any connector or remove or replace any components while power is applied to equipment.



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b. Extreme care must be taken to prevent entrance of noncontained fuel, or other foreign substances into the compartment in which the equipment is installed.

3.2.7 Input requirements.

3.2.7.1 Electrical. The Receiver, Radio R-1963/ARN has been designed to provide reliable performance when supplied with electrical input power with characteristics as defined in MIL-STD-704 and limits as specified below:

a. DC Power

| | |
|---|---|
| Voltage limits | 24.75 to 30.25 volts |
| Power | 7.5 watts max. (at 30.25 volts) |

3.2.8 Environmental requirements. The Receiver, Radio R-1963/ARN components comply with the environmental requirements specified in 3.2.14.2. Components of the Receiver, Radio R-1963/ARN shall be installed within the limits specified.

3.2.9 Equipment heat dissipation. Heat dissipation data for each unit of equipment is given in Table 1.

3.2.10 Total weight. The total weight of the equipment exclusive of cables is 4.1 pounds (maximum). The weight and center of gravity of the individual units is as specified in Table I and Figures 4 and 5 respectively. (See Appendix)

3.2.11 Location considerations. Every consideration shall be given in the location of equipment and in the design of installation details to promote operator efficiency and ease of adjustment and replacement. These considerations shall not be compromised to facilitate aircraft production techniques.

3.2.12 Pressurization. Pressurization of the Receiver, Radio R-1963/ARN is not required.

3.2.13 Cooling. No special cooling provisions are required for the Radio Receiver R-1963/ARN.

3.2.14 Detailed requirements.

3.2.14.1 Installation requirements (See Figures 2 thru 5). (See Appendix)

3.2.14.1.1 MT-4835/ARN Mounting Base, Electrical Equipment (See Figure 5). The Mounting Base, Electrical Equipment MT-4835/ARN, hereafter referred to in this paragraph as the mount, shall be installed in a location where the ambient environmental conditions do not exceed those specified in 3.2.14.2. In addition, the area in which the mount is installed shall provide sway space for the R-1963/ARN receiver (when installed in mount), and clearance for removal of the receiver from the mount and connection of mating cables, as shown in Figure 5.



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The mounting (hole) pattern for the mount is also shown in Figure 5. All mounting surfaces shall be free of oxides, dirt, oil, paint, grease, etc. and prepared and treated in accordance with MIL-B-5087 prior to securing the mount in place. The four bonding straps (one provided for each vibration isolator) shall be installed between the bottom of the vibration isolator and the mounting surface of the aircraft; movement of the floating portion of the mount shall not be impaired by the bonding straps. Four #8-32 screws shall be used in each vibration isolator to secure the mount in place.

3.2.14.1.2 Receiver, Radio R-1963/ARN (See Figure 4). The Receiver, Radio R-1963/ARN, hereafter referred to in this paragraph as the receiver, shall be installed in the Mounting Base, Electrical Equipment MT-4835/ARN and in the location described in 3.2.14.1.1. Installation procedure for installing the receiver into the mount is as follows:

a. Loosen knurled adjustment screw located on front of MT-4835/ARN.

b. Slide receiver into mounting tray of MT-4835/ARN until angled flange located on rear of receiver engages with mounting lip located on rear of mounting tray,

c. Tighten knurled adjustment screw on MT-4835/ARN until hold-down clamp is firmly seated over mounting flange located on front of receiver.

d. Check that sway space around the top, sides, rear and front of the receiver meet the limits shown in Figure 5.

3.2.14.2 Environmental limitations.

3.2.14.2.1 Temperature. Non-operating temperature range shall be -54°C to +71° C. Short time (30 minutes maximum) operating high temperature shall be +71°C. Normal operating temperature range shall be -46°C to +55°C.

3.2.14.2.2 Altitude. Maximum operating altitude without degradation in specified performance shall be up to 30,000 feet above sea level. Maximum transporting altitude without degradation of specified performance shall be up to 50,000 feet above sea level,

3.2.14.2.3 Humidity. The equipment is capable of operation without degradation in specified performance during and after prolonged exposure to humidity levels up to 95%.

3.2.14.2.4 Shock.

3.2.14.2.4.1 Equipment. Shall not exceed 3 shocks of 6 g's; each shock impulse shall not exceed 11 ± 2 milliseconds in time duration.

3.2.14.2.4.2 Mounting base (crash safety shock). There shall be no fialure of the mounting base, isolators, etc. and the receiver shall remain in place if a shock of 15 g's for 11 ± 2 milliseconds is applied.



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3.2.14.2.5 Explosion. There shall be no ignition of an ambient explosive gaseous mixutre with air when operating in such an atmosphere.

3.2.14.2.6 Vibration. Maximum vibration when the equipment is normally mounted with vibration isolators in place, shall not exceed the vibration frequency and amplitude range shown in curve M, method 514.1 of MIL-STD-8lCB.

3.3 Bench test requirements. Unless otherwise specified, the applicable tests in section 4 shall be conducted on all equipments comprising the Receiver, Radio R-1963/ARN to insure compliance with the following requirements. Compliance with the specified bench test requirements shall be verified prior to installation in an aircraft.

3.3.1 Test bench (See 4.5.1). A permanently located bench or test rack shall be provided for mounting the Receiver, Radio R-1963/ARN components and specified test equipment. The arrangement shall provide easy access to the component parts and adjustments. The test rack shall be completely cabled to simulate the normal aircraft installation. The input power source shall be capable of providing power within the limits specified in 3.2.7.

3.3.2 Equipment condition (See 4.5.2). The equipment shall be free of dust, dirt, foreign or packing material and any external or internal damage.

3.3.3 Glideslope receiver section (See 4.5.3).

3.3.3.1 Receiver sensitivity (See 4.5.3.1). with a 25 microvolt standard glideslope deviation signal input, the glideslope deviation current into a 1000 ohm load shall be 47 to 95 microampere (standard glideslope deflection) and the glideslope warning flag current into a 1000 ohm load shall be 235 microampere or more.

3.3.3.2 AGC characteristics (See 4.5.3.2). With a standard glideslope deviation signal input varying in amplitude from 100 to 10,000 microvolt, the glide slope deviation current into a 1000 ohm load shall not vary more than ±11 rnicroamperes from the deviation current obtained with a 700 microvolt input signal.

3.3.3.3 Course centering (See 4.5.3.3). with a 700 microvolt standard glideslope centering signal input, the glideslope deviation current into a 1000 ohm load shall not exceed ±15 microampere.

3.3.3.4 Deflection sensitivity (See 4.5.3.4). With 700 microvolt standard glideslope deviation signal input, the glideslope deviation current into a l000 ohm load shall be 60 to 95 microampere (standard deflection) when the 90 Hz tone predominates. When the 150 Hz tone predominates the glideslope deviation current into a 1000 ohm load shall be -60 to -95 microampere (standard deflection).

3.3.3.5 Warning flag. (See 4.5.3.5). With a 700 microvolt standard glideslope centering signal input, the glideslope warning flag current into a 1000 ohm load shall be not less than 235 microampere. With a 700 microvolt standard glideslope



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centering signal input, the glideslope flag current shall be 180 microampere or less when any of the following conditions exist:

a. No rf signal present on channel frequency selected.

b. 90 Hz or 150 Hz modulation absent from detected signal.

c. Percentage of modulation of the 90 Hz and 150 Hz signal is reduced to 50% of its nominal level (or 20% modulation).

3.3.4 Marker beacon receiver section (See 4.5.4).

3.3.4.1 Receiver threshold sensitivity (See 4,5.4.1). With a 75 MHz input signal modulated 95% at 1300 Hz, the input signal level required to produce a MB lamp response of half brilliance shall be between 350 to 700 microvolt in the high sensitivity mode of operation and between 1000 to 2100 microvolt in the low sensitivity mode of operation.

3.3.4.2 Audio output power (See 4.5.4.2). With a 75 MHz input signal modulated 95% at 1300 Hz, the input signal level required to produce an audio output between 40 and 200 milliwatts into a 150 ohm load shall be 10 times, or less, the input signal level which produces a MB lamp response of half brilliance.

3.3.4.3 Lamp actuation (See 4.5.4.3). With a 75 MHz, 50,000 microvolt input signal, three separate lamp driving outputs shall be provided with only one output actuated for each input signal, 95% modulation frequency of 400, 1300 and 3000 Hz.

3.4 Engineering approval requirements. Engineering approval is required prior to final acceptance by the procuring activity of the Receiver, Radio R-1963/ARN installation and operation. The applicable tests specified in section 4 shall be conducted on a prototype or first production model aircraft of a given contract and on the first aircraft following a major change in either the aircraft or the equipment installation design to insure compliance with the engineering approval requirements specified herein. The aircraft selected shall be fully configured (electronics, armament, armor plating, etc.) as specified in the aircraft contract unless otherwise approved by the procuring activity. If the installation is part of a retrofit program, all applicable MWO's shall be applied prior to conducting the engineering approval tests. Engineering approval requirements shall consist of the preflight, flight and E MC requirements specified in 3.4.1 through 3.4.3.

3.4.1 Preflight. Compliance with the following preflight requirements shall be verified prior to determining compliance with the flight requirements specified in 3.4.2 and after verification of the bench test requirements specified in 3.3. Engineering preflight tests specified in 4.6.1 shall be performed with Receiver, Radio R-1963/ARN installed in the aircraft in accordance with 3.2. The tests specified in 4.6.1 shall be conducted at a location which is free of electromagnetic energy reflecting surfaces such as buildings, other aircraft, etc. and where the electromagnetic environment level is not more than 4 dB above the receider internal background level. Any test site which does not meet these requirements' shall not be utilized without prior approval of the procuring activity.



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3.4.1.1 Electrical power source (See 4.6.1.1). Electrical power shall be obtained from the aircraft primary dc bus. The supply voltage measured at the input terminals of the Receiver, Radio R-1963/ARN shall be 24.75 to 30.25 volts dc.

3.4.1.2 Glideslope operation (See 4.6.1.2).

3.4.1.2.1 Receiver sensitivity (See 4.6.1.2.1). With a 25 microvolt standard glideslope deviation signal (90 Hz predominant) appearing at the glideslope antenna input, the glideslope deviation needle shall deflect down and situate between the first and third deviation dots (standard deflection) on the ID-1347C/ARN-82 and the glide-slope warning flag shall be out of view.

3.4.1.2.2 Course centering See 4.6.1.2.2). with a 700 microvolt standard glideslope centering signal appearing at the glideslope antenna input, the glideslope deviation needle shall be within the center circle of the ID-1347C/ARN-82.

3.4.1.2.3 Deflection sensitivity (See 4.6.1.2.3). With a 700 microvolt standard glideslope deviation signal appearing at the glideslope antenna input, the glideslope deviation needle shall deflect up and situate between the first and third deviation dots (standard deflection) on the ID-1347C/ARN-82 when the 150 Hz tone predominates. When the 90 Hz tone predominates, the glideslope deviation needle shall deflect down and situate between the first and third deviation dots (standard deflection) on the ID-1347C/ARN-82.

3.4.1.2.4 Warning flag (See 4.6.1.2.4). With a 700 microvolt standard glide-slope centering signal at the glideslope antenna input, the glideslope warning flag shall be out of view. The glideslope warning flag shall be in view when any of the following conditions exist.

a. No rf signal present on channel frequency selected.

b. 90 Hz or 150 Hz modulation absent from detected signal.

c. Percentage of modulation of the 90 Hz and 150 Hz signal is reduced to 50% of its nominal level (or 20% modulation).

3.4.1.3 Marker beacon operation (See 4.6.1.3).

3.4.1.3.1 Threshold sensitivity (See 4.6.1.3.1). An audio output shall be evidenced in the HI sensitivity mode of operation when the MB volume knob is fully clockwise and a 700 microvolt, 75 MHz marker signal modulated 95% by 1300 Hz is applied to the marker antenna input. An audio output shall be evidenced in the LO sensitivity mode of operation when the MB volume knob is fully clockwise and a 2100 microvolt, 75 MHz marker signal modulated 95% by 1300 Hz is applied to the marker antenna input.

3.4.1.3.2 Audio output power (See 4.6.1.3.2). Marker audio output shall be variable and intelligible over a signnal range from receiver threshold to 10 times receiver threshold at marker antenna input.



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3.4.1.3.3 Lamp actuation (See 4.6.1.3.3). With an rf signal of 75 MHz, at 50,000 microvolt applied to marker antenna input, three separate lamp driving outputs shall be provided with only one output actuated for each input signal, 95% modulation frequency of 400, 1300, and 3000 Hz. With Marker Test switch energized, three lamp driving outputs shall be provided.

3.4.1.4 Glideslope system operation (See 4.6.1.4). Using a portable glideslope ramp test transmitter producing approximately 700 microvolt standard glideslope centering and variable glideslope deviation signals at the aircraft antenna output, the Receiver, Radio R-1963/ARN shall demonstrate the following:

a. With a standard glideslope centering signal, the glideslope deviation needle shall be within the center circle.

b. With a glide slope signal varying from 0 DDM to 0.175 DDM, the glide-slope deviation needle shall vary from center at 0 DDM to between the third and fourth deviation dots (maximum deflection) at 0.175 DDM.

c. The glideslope warning flag shall be out of view until the 90 Hz and 150 Hz modulating tone is removed, at which time the glideslope warning flag shall come completely into view.

3.4.1.5 Marker beacon system operation (See 4.6.1.5). Using a portable marker beacon ramp test transmitter producing an approximate 1250 microvolt signal at the aircraft marker beacon antenna output with 95% modulation at 400, 1300 and 3000 Hz, the Receiver, Radio R-1963/ARN shall illuminate only one of the three marker lamps that is related to the modulation frequency. The single marker lamp shall light in single marker lamp systems at each of the three modulation frequencies.

3.4.2 Flight See 4.6.2). Compliance with the following flight requirements shall be verified after compliance with the preflight requirements specified in 3. 4.1 has been verified. Engineering flight tests specified in 4.6.2 shall be performed to insure the Receiver, Radio R-1963/ARN performance is in accordance with the following flight requirements while all the “normal” in-flight equipments in the air-craft are functioning.

3.4.2.1 Glideslope operation (See 4.6.2.1). Using an operating, properly FAA certified ILS ground station with known and calibrated antenna radiation character-istics, the Receiver, Radio R-1963/ARN and the glideslope antenna system shall demonstrate the flight requirements specified in 3.4.2.1.1 and 3.4.2.1.2.

3.4.2.1.1 Receiver, Radio R-1963/ARN glideslope flight requirements (See 4.6.2.1.1).

a. Glideslope deviation needle shall be centered when flying on the glide-slope path.

b. Glideslope deviation needle shall deflect above center position when aircraft is flown below glideslope path.



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c. Glideslope deviation needle shall deflect below center position when aircraft is flown above glieslope path.

d. Glideslope warning flag shall be out of view when proper glideslope signal is received and completely in view when a proper glideslope signal is not received.

3.4.2.1.2 Glideslope antenna field pattern and system sensitivity flight requirements (See 4.6.2.1.2). The glideslope antenna system shall provide sufficient signal to the glideslope receiver to cause the glideslope warning flag to be out of view when the aircraft is on the glideslope path, approximately centered on the localizer course, at ranges up to at least 15 nautical miles from the approach end of the runway, at all normal approach attitudes and with the direction to the glideslope station at all angles between 30° left and 30° right of the aircraft nose.

3.4.2.2 Marker beacon operation (See 4.6.2.2). Using operating, properly FAA certified marker beacons with known and calibrated antenna radiation patterns, the Receiver, Radio R-1963/ARN and marker beacon antenna system shall demonstrate the flight requirements specified in 3.4.2.2.1 and 3.4.2.2.2.

3.4.2.2.1 Receiver, Radio R-1963/ARN marker beacon flight requirements (See 4.6.2.2.1). When flying the localizer inbound course over ILS marker beacons the marker beacon operation shall be as follows:

a. Blue marker lamp (or single marker lamp) shall light when passing over the outer marker; 400 Hz tone shall be audible.

b. Amber marker lamp (or single marker lamp) shall light when passing over middle marker; 1300 Hz tone shall be audible.

c. White marker lamp (or single marker lamp) shall light when passing over inner marker; 3000 Hz tone shall be audible.

3.4.2.2.2 Marker beacon antenna pattern and system sensitivity flight requirements (See 4.6.2.2.2). The marker beacon antenna pattern shall provide sufficient signal to the marker beacon receiver, to present the proper visual and audio indications, when the aircraft is flying in normal enroute and approach flight attitudes, at altitudes up to 10,000 feet AGL, over marker beacons.

3.4.2.3 ILS flight checks (See 4.6.2.3). During the execution of several ILS approaches, using an operating, properly FAA certified ILS ground station with known and calibrated antenna radiation patterns and known distance, the Receiver, Radio R-1963/ARN shall demonstrate the following.

a. Glideslope warning flag shall be out of view to indicate normal operation.

b. Proper aircraft altitude shall be indicated at the outer marker and middle marker when tracking the glideslope path.

c. Correct tracking of the glideslope path shall be evidenced.



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d. Correct marker indications shall be received when passing over the marker beacons. The marker signals shall be received in the following sequence:

| Marker Beacon | Light Color | Audio Tone Frequency |
|---|---|---|
| Outer | Blue | 400 Hz |
| Middle | Amber | 1300 Hz |
| Inner | White | 3000 Hz |

In single marker lamp systems only the green marker lamp shall illuminate when passing over the marker beacons.

3.4.3 Electromagnetic compatibility (See 4.3). Compliance with the complete functional electrical-electronic requirements specified in the procuring activity approved, contractor prepared EMC control and EMC test plans required in 3.1.11 shall be verified.

3.5 Production approval requirements. production approval is required prior to final acceptance by the procuring activity of the Receiver, Radio R-1963/ARN installation and operation. The applicable tests specified in section 4 shall be conducted on every aircraft submitted for acceptance to insure compliance with the production approval requirements specified herein. The aircraft shall be fully configures (electronics, armament, armor plating, etc.) as specified in the aircraft contract unless otherwise approved by the procuring activity. If the installation is part of a retrofit program, all applicable MWO's shall be applied prior to conducting the production approval tests. Production approval requirements shall consist of the preflight, flight and EMC requirements specified in 3.5.1 through 3.5.3.

3.5.1 Preflight. Compliance with the following preflight requirements shall be verified prior to determining compliance with the flight requirements specified in 3.5.2 and after verification of the bench test requirements specified in 3.3. Production preflight tests in 4.7.1 shall be performed with Receiver, Radio R-1963/ARN installed in the aircraft in accordance with 3.2 to insure the equipment installation and operation is in accordance with the following preflight requirements. The tests specified in 4.7.1 shall be conducted at a location which is free of electromagnetic energy reflecting surfaces such as buildings, other aircraft, etc. and where the electromagnetic environment level is not more than 4 dB above the receiver internal background level. Any test site which does not meet these requirements shall not be utilized without prior approval of the procuring activity.

3.5.1.1 Electrical power source (See 4.7.1.1). The supply voltage measured at the power input terminals of the Receiver, Radio R-1963/ARN shall be 24.75 to 30.25 volts dc.



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3.5.1.2 Glideslope operation (See 4.7.1.2). Using a portable glideslope ramp test transmitter producing approximately 700 microvolt standard glideslope centering and variable glideslope deviation signals at the aircraft glideslope-antenna input, the Receiver, Radio R-1963/ARN shall demonstrate the following:

a. With a standard glideslope centering signal, the glideslope deviation needle shall be within the center circle of the ID-1347C/ARN-82.

b. With a glideslope signal varying from 0 DDM to 0.175 DDM, the glideslope deviation needle shall vary from center at 0 DDM to between the third and fourth deviation dots (maximum deflection) on the ID-1347C/ARN-82 at 0.175 DDM.

c. The glideslope warning flag shall be out of view until the 90 or 150 Hz modulating tone is removed, at which time the glideslope warning flag shall come completely into view.

3.5.1.3 Marker beacon operation (See 4.7.1.3). Using a portable marker beacon ramp test transmitter producing an approximate 1250 microvolt signal at the aircraft marker beacon antenna output with 95% modulation at 400, 1300, and 3000 Hz, the Receiver, Radio R-1963/ARN shall illuminate only the one of three marker lamps that is related to the modulation frequency. The single lamp shall light in single marker lamp systems at each of the three modulation frequencies.

3.5.2 Flight. Compliance with the following flight requirements shall be verified after compliance with the preflight requirements specified in 3.5.1 have been verified. Production flight tests specified in 4.7.2 shall be performed to insure the Receiver, Radio R-1963/ARN performance is in accordance with the following flight requirements while all “normal” in-flight equipments in the aircraft are functioning.

3.5.2.1 Glideslope operation (See 4.7.2.1). Using an operating, properly FAA certified, ILS ground station with known and calibrated antenna radiation pattern and known distance, the Receiver, Radio R-1963/ARN shall demonstrate the following:

a. Glideslope warning flag shall be out of view at least 15 nautical miles from the ILS ground station when flying the localizer inbound course.

b. Glideslope deviation needle shall be centered when flying on the glideslope path.

c. Glideslope deviation needle shall deflect above center position when aircraft is flown below glideslope path.

d. Glideslope deviation needle shall deflect below center position when aircraft is flown above glideslope path.



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3.5.2.2 Marker beacon operation (See 4.7.2.2). Using an operating, properly FAA certified ILS ground station with known and calibrated antenna radiation patterns, the Receiver, Radio R-1963/ARN shall demonstrate the following:

a. Blue marker lamp (or single marker lamp) shall light when passing over the outer marker; 400 Hz tone shall be audible.

b. Amber marker lamp (or single marker lamp) shall light when passing over the middle marker; 1300 Hz tone shall be audible.

c. White marker lamp (or single marker lamp) shall light when passing over the inner marker; 3000 Hz tone shall be audible.

3.5.2.3 ILS flight checks (See 4.7.2.3). During execution of several ILS approaches, using an operating, properly FAA certified ILS ground station with known and calibrated antenna radiation patterns and known distance, the Receiver, Radio R-1963/ARN shall demonstrate the following:

a. Glideslope warning flag shall be out of view to indicate normal operation.

b. Proper aircraft altitude shall be indicated at the outer marker and middle marker when tracking the glideslope path.

c. Correct tracking of the glideslope path shall be evidenced.

d. Correct marker indications shall be received when passing over the marker beacons. The marker signals shall be received in the following sequence:

| Marker Beacon | Light Color | Audio Tone Frequency |
|---|---|---|
| Outer | Blue | 400 Hz |
| Middle | Amber | 1300 Hz |
| Inner | White | 3000 Hz |

In single marker lamp systems only the green marker lamp shall illuminate when passing over the marker beacons.

3.5.3 Electromagnetic compatibility (See 4.3). Compliance with the limited functional electrical-electronic requirements specified in the procuring activity approved, contractor prepared EMC control and EMC test plan required in 3.1.11 shall be verified.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial facility acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. The examinations and tests required to assure conformance with the requirements of section 3 are classified as follows :

a. Electromagnetic compatibility tests

b. Installation examination

c. Bench tests

d. Engineering approval tests

e. Production approval tests

4.3 Electromagnetic compatibility tests. Under the test conditions and procedures specified in the procuring activity approved, contractor prepared EMC control and EMC test plans, demonstrate compliance with 3.4.3 and 3.5.3.

4.4 Installation examination. All Receiver, Radio R-1963/ARN electronic equipment, whether Government furnished or contractor furnished, which is installed by the contractor shall be examined to determine compliance with 3.2.

4.5 Bench tests. Prior to performing the glideslope-marker beacon bench test procedures, the controls of Radio Test Set AN/ARM-92B should be set as follows:

| | |
|---|---|
| OFF/PWR/TEST | OFF |
| MODE | 400 Hz |
| BEARING control | 0 |
| BEARING switch | BRG |
| FLAG LOAD | 4 |
| SENS | LO |
| IND TEST | GS RCVR |
| DEVIATION | 0 |



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| | |
|---|---|
| CONTROL frequency | 109.30 |
| AUDIO | MB |

4.5.1 Input power. All power sources shall be in compliance with 3.2.7.

4.5.2 Visual and mechanical inspection. All equipment shall be inspected for external damage. Remove all dust, dirt, foreign or packing material. Examine for foreign material and loose or damaged parts for compliance with 3.3.2. Clean, tighten, repair and report in accordance with 3.1.7.

4.5.3 Glideslope receiver section.

4.5.3.1 Receiver sensitivity. Using test setup shown in Figure 6, check receiver sensitivity, as specified below, for compliance with 3.3.3.1. (See Appendix)

a. Set frequency selectors on control panel of AN/ARM-92B Radio Test Set to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 25 microvolt standard glideslope deviation signal at 332.00 MHz with 90 Hz tone predominant. Deviation meter on AN/ARM-92B shall indicate between 47 and 95 microvolt to the left of 0 (zero), and flag meter shall indicate 235 microampere or more.

c. Repeat steps a. and b. at each of the frequency settings given in Table IV.

TABLE IV

CONTROL PANEL of AN/ARM-92B and HP-8640B FREQUENCY SETTINGS

| CONTROL PANEL FREQ. (MHz) | HP-8640B SIG. GEN. FREQ. (MHz) |
|---|---|
| 108.95 | 329.15 |
| 111.35 | 332.15 |
| 110.30 | 335.00 |

4.5.3.2 AGC characteristics. Using test setup shown in Figure 6, check AGC response, as specified below, for compliance with 3.3.3.2.

a. Set frequency selectors on control panel of AN/ARM-92B Radio Test Set to 109.30 MHz.



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b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope deviation signal at 332.00 MHz with 90 Hz tone predominant. Deviation meter on AN/ARM-92B shall indicate between 60 and 95 microampere to the left of 0 (zero). Note deviation indication.

c. Decrease output level of HP-8640B to 100 microvolt, then gradually increase output level to 10,000 microvolt. Deviation meter shall indicate within ±11 microampere of the deviation indication noted in step b. throughout entire output level range.

4.5.3.3 Course centering. Using test setup shown in Figure 6, check course centering, as specified below, for compliance with 3.3.3.3.

a. Set frequency selectors on control panel of AN/ARM-92B Radio Test Set to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32 ILS Audio Generator for a 700 microvolt, standard glideslope centering a signal at 332.00 MHz. Deviation meter on AN/ARM-92B shall indicate 0 ± 15 microampere.

4.5.3.4 Deflection sensitivity. Using test setup shown in Figure 6, check deflection sensitivity, as specified below, for compliance with 3.3.3.4.

a. Set frequency selectors on control panel of AN/ARM-92B Radio Test Set to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope deviation signal at 332.00 MHz with 90 Hz tone predominant. Deviation meter on AN/ARM-92B shall indicate between 60 and 95 microampere to the left of 0 (zero).

c. Set RIG-32A for a standard glideslope deviation signal with 150 Hz tone predominant. Deviation meter on AN/ARM-92B shall indicate between 60 and 95 microampere to the right of 0 (zero).

4.5.3.5 Warning flag. Using test setup shown in Figure 6, check glideslope warning flag operation, as specified below, for compliance with 3.3.3.5.

a. Set frequency selectors on control panel of AN/ARM-92B Radio Test Set to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope centering signal at 332.00 MHz. Flag meter on AN/ARM-92B Radio Test Set shall indicate 235 microampere or more.

c. Set RIG-32A output for 90 Hz modulating tone only. Flag meter shall indicate less than 180 microampere.



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d. Set RIG-32A output for standard glideslope centering signal. Flag meter shall indicate 235 microampere or more.

e. Set RIG-32A output for 150 Hz modulating tone only. Flag meter shall indicate less than 180 microampere.

f. Set RIG-32A output for standard glideslope centering signal, then reduce modulation level of 90 and 150 Hz signal by 50% (20% modulation). Flag meter shall indicate less than 180 microampere.

g. Increase modulation level of 90 and 150 Hz tones of RIG-32A to 40%. Flag meter shall indicate 235 microampere or more.

4.5.4 Marker beacon receiver section

4.5.4.1 Receiver threshold sensitivity. Using test setup shown in Figure 7, check receiver threshold sensitivity, as specified below, for compliance with 3.3.4.1. (See Appendix)

a. Place HI/LO sensitivity switch on AN/ARM-92B Radio Test Set to HI.

b. Adjust HP-8640B Rf Signal Generator for an output signal of 75 MHz, modulated 95% by 1300 Hz, then adjust rf output level to produce a MB lamp response of half brilliance. The output level of HP-8640B shall be 350 to 700 microvolt.

c* Place HI/LO sensitivity switch to LO.

d. Increase output level of HP-8640B to produce a MB lamp response of half brilliance. The output level of HP-8640B shall be 1000 to 2100 microvolt.

4.5.4.2 Audio output power. Using test setup shown in Figure 7, check audio output power, as specified below, for compliance with 3.3.4.2.

a. Place HI/LO sensitivity switch on AN/ARM-92B Radio Test Set to LO.

b. Adjust HP-8640B Rf Signal Generator for an output signal of 75 MHz, modulated 95% by 1300 Hz then adjust output level to produce a MB lamp response of half brilliance. Note rf output level.

c. Increase output level of HP-8640B to 10 times that noted in step b above. The reading on TS-585( )/U shall be between 40 and 200 milliwatts.

4.5.4.3 Lamp actuation. Using test setup shown in Figure 7, check lamp actuation, as specified below for compliance with 3.3.4.3.

a. Set HP-8640B Rf Signal Generator for an output signal of 75 MHz, modulated 95% by 400 Hz at 50,000 microvolt. The blue (400 Hz) marker lamp on AN/ARM-92B Radio Test Set shall be illuminated and the amber and white marker lamps shall be extinguished.



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b. Increase modulation frequency of HP-8640B to 1300 Hz while maintaining 95% modulation. The amber (1300 Hz) marker lamp on AN/ARM-92B shall be illuminated and the blue and white marker lamps shall be extinguished.

c. Increase modulation frequency of HP-8640B to 3000 Hz while maintaining 95% modulation. The white (3000 Hz) marker lamp on AN/ARM-92B shall be illuminated and the blue and amber marker lamps shall be extinguished.

4.6 Engineering approval tests.

4.6.1 Preflight tests.

4.6.1.1 Electrical power source. With all electrical equipment which is connetted to the same bus as the Receiver, Radio R-1963/ARN turned ON and operating in a normal manner, place the Receiver, Radio R-1963/ARN in the ON mode of operation. Under this condition, to determine compliance with 3.4.1.1, measure the dc input voltage level at pin 5 (+) and pin 19 (-) of the mating connector (Pi) for the Radio Receiver, R-1963/ARN with HP-410C Vacuum Tube Voltmeter. The HP-41OC shall read between 24.75 and 30.25 volts dc.

4.6.1.2 Glideslope operation.

4.6.1.2.1 Receiver sensitivity. Using test setup shown in Figure 8, check receiver sensitivity, as specified below, for compliance with 3.4.1.2.1. (See Appendix)

a. Set frequency selectors on C-6873B/ARN-82 Radio Set Control to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 25 microvolt standard glideslope deviation signal at 332.00 MHz where 90 Hz tone predominates. The glideslope deviation needle on ID-1347C/ARN-82 Course Deviation Indicator shall deflect down and situate between the first and third deviation dots and the glide slope warning flag shall be out of view.

c. Repeat steps a. and b. at each frequency listed in Table IV.

4.6.1.2.2 Course centering. Using test setup shown in Figure 8, check course centering, as specified below, for compliance with 3.4.1.2.2.

a. Set frequency selectors on C-6873B/ARN-82 Radio Set Control to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope centering signal at 332.00 MHz. The glideslope deviation needle on ID-1347C/ARN-82 Course Deviation Indicator shall be within the center circle.



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4.6.1.2.3 Deflection sensitivity. Using test setup shown in Figure 8. check deflection sensitivity, as specified below, for compliance with 3.4.1.2.3.

a. Set frequency selectors on C-6873B/ARN-82 Radio Set Control to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope deviation signal at 332.00 MHz with 90 Hz tone predominant. The glideslope deviation needle on ID-1347C/ARN-82 Course Deviation Indicator shall deflect down and situate between the first and third deviation dots.

c. Set RIG-32A for a standard glideslope deviation signal with 150 Hz tone predominant. The glideslope deviation needle on ID-1347C/ARN-82 shall deflect up and situate between the first and third deviation dots.

4.6.1.2.4 Warning flag. Using test setup shown in Figure 8, check the warning flag operation, as specified below, for compliance with 3.4.1.2.4.

a. Set frequency selectors on C-6873B/ARN-82 Radio Set Control to 109.30 MHz.

b. Adjust HP-8640B Rf Signal Generator and RIG-32A ILS Audio Generator for a 700 microvolt standard glideslope centering signal at 332.00 MHz. The glideslope warning flag on ID- 1347C/ARN-82 Course Deviation Indicator shall be out of view.

c. Remove 150 Hz tone from RIG-32A output signal. The glideslope warning flag shall be in view.

d. Set RIG-32A output for a standard glideslope centering signal. The glideslope warning flag shall be out of view.

e. Remove 90 Hz tone from RIG-32A output signal. The glideslope warning flag shall be in view.

f. Set RIG-32A output for a standard glideslope centering signal. The glideslope warning flag shall be out of view.

g. Reduce modulation level of 90 and 150 Hz tone output of RIG-32A by 50% (20% modulation). The glideslope warning flag shall be in view.

h. Increase modulation level of 90 and 150 Hz tone to 40%. The glideslope warning flag shall be out of view.



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4.6.1.3 Marker beacon operation.

4.6.1.3.1 Threshold sensitivity. Using test setup shown in Figure 9, check threshold sensitivity, as specified below, for compliance with 3.4.1.3.1.

a. Connect headset into aircraft intercom system, rotate marker audio volume control fully clockwise, and set HI/LO sensitivity switch to HI.

b. Adjust HP-8640B Rf Signal Generator for an rf output signal of 75 MHz modulated 95% by 1300 Hz at 700 microvolt. An audio tone shall be audible in headset.

c. Place HI/LO sensitivity switch to LO.

d. Increase output level of HP-8640B to 2100 microvolt. An audio tone shall be audible in headset.

4.6.1.3.2 Audio output power. Using test setup shown in Figure 9, check audio output power, as specified below, for compliance with 3.4.1.3.2.

a. Connect headset into aircraft intercom system, rotate marker audio volume control fully clockwise, and set HI/LO sensitivity switch to HI.

b. Adjust HP-8640B Rf Signal Generator for a rf output signal of 75 MHz, modulated 95% by 1300 Hz, then adjust rf output level until an audio tone is evidenced in headset. Note rf output level.

c. Set marker audio volume control to mid-range, then slowly increase rf output level of HP-8640B to 10 times the level noted in step b. Audio shall be audible throughout entire signal input range.

4.6.1.3.3 Lamp actuation. Using test setup shown in Figure 9, check lamp actuation, as specified below, for compliance with 3.4.1.3.3.

a. Set HI/LO sensitivity switch to HI.

b. Set HP-8640B Rf Signal Generator for an output signal of 75 MHz, modulated 95% by 400 Hz at 50,000 microvolt. The blue (400 Hz) marker lamp on marker lamp panel (or single marker lamp) shall be illuminated and the amber and white marker lamps shall be extinguished.

c. Increase modulation frequency of HP-8640B to 1300 Hz while maintaining 95% modulation. The amber (1300 Hz) lamp on marker lamp panel (or single marker lamp) shall be illuminated and the blue and white marker lamps shall be extinguished.

d. While maintaining 95% modulation, set HP-8640B to 3000 Hz. The white (3000 Hz) lamp on marker lamp panel (or single marker lamp) shall be illuminated and the blue and amber marker lamps shall be extinguished.



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e. Place function selector on C-6873B/ARN-82 Radio Set Control to TEST. The three marker lamps (or single lamp in single marker lamp installations) shall illuminate.

4.6.1.4 Ramp Test Set operation. Position the ramp test set approximately 20 feet from the appropriate aircraft antenna keeping the test set as near the center line of the aircraft as possible, (The test set may be positioned either in front of or behind the aircraft) and maintain line of sight between ramp test set and aircraft antenna. Conduct tests utilizing 972Q-4 VOR/ILS/MB Ramp Test Set in accordance with the instructions in paragraphs 4.6.1.5 and 4.6.1.6.

4.6.1.5 Glideslope operation. Using the 972Q-4 VOR/ILS/MB Ramp Test Set with antenna, check glide slope operation, as specified below for compliance with 3.4.1.4.

a. Ascertain that glideslope antenna coax cable is connected to the glideslope antenna.

b. Connect the 972Q-4 test set to the test set antenna. Set the 972Q-4 for 30 dB of output attenuation. Adjust the 972Q-4 for a standard glideslope centering signal at 334.70 MHz.

c. Set frequency selectors on C-6873B/ARN-82 Radio Set Control to 108.10 MHz.

d. Move the test set antenna towards the aircraft glideslope antenna until the glideslope warning flag on the ID-1347C/ARN-82 Course Deviation Indicator goes out of view. Set the 972Q-4 for 7 dB of output attenuation (remove 23 dB of attenuation).

e. With the 972Q-4 set for a standard glideslope centering signal, the glideslope deviation needle on the ID-1347C/ARN-82 shall be within the center circle.

f. Adjust the 972Q-4 for a variable glideslope deviation signal. Vary the glideslope deviation signal from 0 DDM to 0.175 DDM, 90 Hz predominating; the glideslope deviation needle on the ID-1347C/ARN-82 shall deflect from center (at 0 DDM) then down to between the third and fourth deviation dots (at 0.175 DDM) with no hang-ups.

g. Vary the glideslope deviation signal from 0 DDM to 0.175 DDM, 150 Hz predominating; the glideslope deviation needle on the ID-1347C/ARN-82 shall deflect from center (at 0 DDM) then up to between the third and fourth deviation dots (at 0.175 DDM) with no hang-ups.

h. Adjust the 972Q-4 to remove the 150 Hz modulation; the glideslope warning flag on the ID-1347C/ARN-82 shall come into view.



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4.6.1.6 Marker beacon operation. Using the 972Q-4 VOR/ILS/MB Ramp Test Set with antenna, check marker beacon operation, as specified below for compliance with 3.4.1.5.

a. Ascertain that marker beacon antenna coax cable is connected to marker beacon antenna.

b. Connect the 972Q-4 test set to the test set antenna.

c. Connect headset into aircraft intercom system.

d. Set the MB SENS switch to HI position and the MB VOL control fully clockwise.

e. With the 972Q-4 located as specified in paragraph 4.6.1.4, set the 972Q-4 controls for an output signal of 75 MHz, modulated 95% by 400 Hz. Move the 972Q-4 towards the aircraft until a strong audio tone is audible. The blue marker lamp (or single marker lamp as applicable) on marker lamp panel shall be illuminated.

f. Set tone modulation output of 972Q-4 to 1300 Hz. The amber marker lamp (or single marker lamp as applicable) shall be illuminated and 1300 Hz tone shall be audible.

g. Set tone modulation output of 972Q-4 to 3000 Hz. The white marker lamp (or single marker lamp as applicable) shall be illuminated and 3000 Hz tone shall be audible.

4.6.2 Flight tests.

4.6.2.1 Glideslope operation. The glide slope operation test shall be conducted with the C-6873B/ARN-82 controls set for the ILS mode of operation. The test site shall be over level terrain and shall include operating, properly FAA certified glide-slope and localizer ground stations of known position and known-and-calibrated antenna radiation pattern. The instructions given in 4.6.2.1.1 and 4.6.2.1.2 shall apply when conducting the test to determine compliance with 3.4.2.1.1 and 3.4.2.1.2, respectively.

4.6.2.1.1 Receiver, Radio R-1963/ARN glideslope flight requirements.

a. Fly toward the airport on the glide slope path; the glide slope warning flag shall be out of view.

b. With the glideslope deviation needle centered, fly down slightly; the glideslope deviation needle shall deflect up. Fly up past the imaginary glidesIope center line; the glideslope deviation needle shall pass through center position to a down deflection.



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4.6.2.1.2 Glideslope antenna field pattern ane system sensitivity flight requirements.

a. Determine the altitude at which the glideslope path intercepts a point, on the localizer center line, at least 15 nautical miles from the approach end of the runway.

b. While flying at that altitude, establish an inbound course to pass through the 15 nautical mile point at approximately a 30° right angle to the localizer center line.

c. The fly-through of the localizer course shall begin at a point far enough beyond the 15 nautical mile point, but still within the localizer sector, to give a fly-up deflection of the glideslope deviation needle.

d. Normal glideslope deviation needle action shall be from fly-up to fly-down while passing through the glideslope path.

e. The glideslope warning flag shall remain out of view throughout the test pattern.

f. Repeat steps b. thru e. except fly the aircraft at approximately a 30° left angle to the localizer center line.

4.6.2.2 Marker beacon operation. The marker beacon operation test shall be conducted with the C-6873B/ARN-82 controls set for the ILS mode and marker beacon low sensitivity operation (unless otherwise indicated). The test site shall be over level terrain and shall include operating, properly FAA certified marker beacons and localizer ground stations of known position and known-and-calibrated antenna radiation patterns. The instructions given in 4.6.2.2.1 and 4.6.2.2.2 shall apply when conducting the test to determine compliance with 3.4.2.2.1 and 3.4.2.2.2.

4.6.2.2.1 Receiver, Radio R-1963/ARN marker beacon flight requirements.

a. Fly the aircraft on a localizer inbound course over the outer, middle, and inner (if available) marker beacons.

b. The marker lamp associated with the beacon being flown over, and only that lamp shall light. The tone associated with the beacon being flown over, and only that tone, shall be audible. Lamps and tones associated with the ILS marker beacons are indicated below. In installations where only a single marker lamp exists, the lamp shall light each time the aircraft flies over an ILS marker beacon.

| MARKER BEACON | LAMP | AUDIO TONE FREQUENCY |
|---|---|---|
| Outer | Blue | 400 Hz |
| Middle | Amber | 1300 Hz |
| Inner | white | 3000 Hz |



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4.6.2.2.2 Marker beacon antenna pattern and system sensitivity flight requirements.

a. Set the MB SENS switch to the HI sensitivity position.

b. Fly the marker beacon test pattern shown in Figure 10, on an airway course, over a low-powered fan marker (class LFM) with a rated power output of 5 watts maximum. (See Appendix)

c. The distance between the point the marker beacon visual and audio indications are first received and the point directly over the marker beacon shall be approximately equal to the distance between the point directly over the marker beacon and the point the marker beacon visual and audio indications are no longer received, for each of the four flight passes.

d. The total distance that the marker beacon visual and audio indications are received, for each of the two flight passes at 10,000 feet, shall not be less than one nautical mile.

e. At 2,000 ft. fly the marker beacon test pattern shown in Figure 11, in a direction which parallels the airway course, over a low-powered fan marker (class LFM) with a rated output of 5 watts maximum. (See Appendix)

f. The applicable marker beacon audio and lamp indications shall be received for approximately the same distance for each of the two flight passes.

Set the MB SENS switch to the LO sensitivity position.

Fly at an altitude of 1000 feet AGL, on a localizer inbound course, with normal stores and landing gear extended, over several ILS marker beacons.

i. The applicable marker beacon lamp and audio indications shall be observed for a distance of not less than 2,000 feet, and not more than 3,000 feet, when the aircraft passes over the ILS marker beacons.

4.6.2.3 ILS flight check. The ILS flight check shall be conducted over level terrain using an operating, properly FM certified ILS ground station with known position and known-and-calibrated antenna radiation patterns. With the C-6873B/ ARN-82 controls set for the ILS mode and low sensitivity marker operation, conduct several ILS approaches. The performance parameters given in 4.6.2.3.1 and 4.6.2.3.2 shall apply during execution of the XIX approaches to determine compliance with 3.4.2.3.

4.6.2.3.1 Marker beacon performance. The correct marker beacon indications (lamp and audio) shall be while passing over the ILS marker beacons.



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4.6.2.3.2 Glideslope performance.

a. The proper altitude shall be indicated on the aircraft barometric altimeter, at the outer and middle marker beacons, while on the glideslope path.

b. The aircraft shall be stabilized on the glideslope path before an altitude of 700 feet above runway elevation is reached.

c. From 700 feet altitude to the decision altitude the pilot shall be able to track the center of the glideslope path to within the first deviation dot on the ID-1347C/ARN-82, or ± 22 feet, whichever is the larger.

d. Flying the centered glideslope deviation needle shall bring the aircraft to the glide slope reference datum of the runway.

4.7 Production approval tests.

4.7.1 Preflight tests.

4.7.1.1 Electrical power source. With all electrical equipment which is connected to the same bus as the Receiver, Radio R-1963/ARN turned ON and operating in a normal manner, place the Receiver, Radio R-1963/ARN in the ON mode of operation. Under this condition, to determine compliance with 3.5.1.1, measure the dc input voltage level at pin 5 (+) and pin 19 (-) of the mating connector (PI) for the Receiver, Radio R-1963/ARN with HP-410C Vacuum Tube Voltmeter. The HP-410C shall indicate 24.75 to 30.25 volts dc.

4.7.1.2 Ramp Test Set operation. Position the ramp test set approximately 20 feet from the appropriate aircraft antenna keeping the test set as near the center line of the aircraft as possible. (The test set may be positioned either i n front of or behind the aircraft) and maintain line of sight between ramp test set and aircraft antenna. Conduct tests utilizing 972Q-4 VOR/ILS/MB Ramp Test Set in accordance with the instructions in paragraphs 4.7.1.3 and 4.7.1.4.

4.7.1.3 Glideslope operation. Using 972Q-4 VOR/ILS/MB Ramp Test Set with antenna, check glide slope operation as specified below, for compliance with 3.5.1.2.

a. Connect the 972Q-4 test set to the test set antenna. Set the 972Q-4 for 30 dB of output attenuation. Adjust the 972Q-4 for a standard glideslope centering signal at 334.70 MHz.

b. Set frequency selectors on C-6873B/ARN-82 Control to 108.10 MHz.

c. Move the test set antenna towards the aircraft glideslope antenna until the glideslope warning flag on the ID-1347C/ARN-82 Course Deviation Indicator goes out of view. Set the 972 Q-4 for 7 dB of output attenuation (remove 23 dB of attenuation).



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d. Set the 972Q-4 for a standard glideslope centering signal; the glideslope deviation needle shall be within the center circle on the ID-1347C/ARN-82.

e. Adjust the 972Q-4 for a variable glideslope deviation signal. Vary the glideslope deviation signal from 0 DDM to 0.175 DDM, 90 Hz predominating; the glideslope deviation needle on the ID- 1347C/ARN-82 shall deflect from center (at 0 DDM) then down to between the third and fourth deviation dots (at 0.175 DDM) with no hang-ups.

f. Vary the glideslope deviation signal from 0 DDM to 0.175 DDM, 150 Hz predominating; the glideslope deviation needle on the ID-1347C/ARN-82 shall deflect from center (at 0 DDM) then up to between the third and fourth deviation dots (at 0.175 DDM) with no hang-ups.

g. Adjust the 972Q-4 to remove the 150 Hz modulation; the glideslope warning flag on the ID-1347C/ARN-82 shall come into view.

4.7.1.4 Marker beacon operation. Using 972Q-4 VOR/ILS/MB Ramp Test Set, check marker beacon operation, as specified below, for compliance with 3.5.1.3.

a. Connect the 972Q-4 test set to the test set antenna.

b. Connect headset into aircraft intercom system.

co Set the MB SENS switch to the LO position and the MB VOL control fully clockwise.

d. With the 972Q-4 located as specified in paragraph 4.7.1.2, set the 972Q-4 controls for an output signal of 75 MHz, modulated 95% by 400 Hz. Move the 972Q-4 towards the aircraft until the blue marker lamp (or single marker lamp as applicable) illuminates (lamp threshold). Move the 972Q-4 back from aircraft until the marker lamp just extinguishes.

e. Place the MB SENS stitch in the HI position; the blue (or single) marker lamp shall illuminate at full brightness and a strong audio tone shall be evidenced.

f. Set tone modulation output of 972Q-4 to 1300 Hz. The amber marker lamp (or single marker lamp as applicable) shall be illuminated and 1300 Hz tone shall be audible.

g. Set tone modulation output of 972Q-4 to 3000 Hz. The white marker lamp (or single marker lamp as applicable) shall be illuminated and 3000 Hz shall be audible.

h. Set function selector switch on C-6873B/ARN-82 to the TEST position. The blue, amber and white marker lamps (or single marker lamp) shall be lit.



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4.7.2 Flight tests.

4.7.2.1 Glideslope operation. The glideslope operation test shall reconducted with the C-6873B/ARN-82 controls set for the ILS mode of operation. The test site shall be over level terrain and shall include an operating, properly FAA certified glideslope ground station of known position and known-and-calibrated antenna radiation fiattern. The following instructions shall apply when conducting the test to determine compliance with 3.5.2.1.

a. Fly toward the airport on the glideslope path; at 15 nautical miles from the glideslope station, the glideslope warning flag shall be out of view.

b. With the glideslope deviation needle centered, fly down slightly; the glideslope deviation needle shall deflect up. Fly up past the imaginary glideslope center line; the glideslope deviation needle shall pass through center position to a down deflection.

4.7.2.2 Marker beacon operation. The marker beacon operation test shall be conducted with the C-6873B/ARN-82 controls set for the ILS mode and marker beacon low sensitivity operation. The test site shall be over level terrain and shall include an operating, properly FAA certified marker beacon and localizer ground station of known position and known-and-calibrated antenna radiation pattern. The following instructions shall apply when conducting the test to determine complimce with 3.5.2.2.

a. Fly the aircraft on a localizer inbound course over the outer, middle, and inner (if available) marker beacons.

b. The marker lamp associated with the beacon being flown over, and only that lamp shall light. The tone associated with the beacon being flown over, and only that tone, shall be audible. Lamps and tones associated with the ILS marker beacons are indicated below. In installations where only a single marker lamp exists the lamp shall illuminate each time the aircraft flies over an ILS marker beacon.

| MARKER BEACON | LAMP COLOR | AUDIO TONE FREQUENCY |
|---|---|---|
| Outer | Blue | 400 Hz |
| Middle | Amber | 1300 Hz |
| Inner | White | 3000 Hz |

4.7.2.3 ILS flight check. the ILS flight check shall be conducted over level terrain using an operating, properly FAA certified ILS ground station with known position and known-and-calibrated antenna radiation patterns. With the C-6873B/ARN-82 controls set for the ILS mode and low sensitivity marker operation, conduct several ILS approaches. The following instructions shall apply when conducting the test to determine compliance with 3.5.2.3.



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a. Fly toward the airport on a localizer inbound course. When the glide-slope path is encountered, begin a landing approach; the glideslope warning flag shall be out of view.

b. The correct marker beacon indications (lamp and audio) shall be received and the proper altitude shall be displayed on the aircraft barometric altimeter while pas sing over the ILS marker beacon.

co From 700 feet altitude above runway elevation to the decision altitude, the pilot shall be able to track the center of the glideslope path to within the first deviation dot on the ID-1347C/ARN-82 or ± 22 feet, whichever is larger.

d. Flying the centered glideslope and VOR/LOC deviation needles shall bring the aircraft to the glideslope reference datum of the runway.



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## 5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.



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6. NOTES

6.1 Intended use. The purpose of this specification is to provide and specify installation instructions and requirements, performance requirements and quality assurance procedures to prohibit serious installation design discrepancies and to insure desirable performance characteristics are achieved when the Receiver, Radio R-1963/ARN is installed in an aircraft.

6.2 Definitions.

6.2.1 Contractor. The word "contractor" as used herein refers to the aircraft manufacturer or other activity responsible for the installation and performance of the quality assurance requirements of Receiver, Radio R-1963/ARN.

6.2.2 Glideslope test signal. The words "glideslope test signal" as used herein refer to an rf carrier at 329.15 to 335.00 MHz amplitude modulated simultaneously with 90 Hz and 150 Hz signals so that the sum of their separate modulation percentage equal 80%.

6.2.3 Standard glide slope centering signal. The words "standard glide slope centering signal" as used herein refer to an rf carrier amplitude modulated simultaneously by 90 Hz (±0.3%) and 150 Hz (±0.3%) signals so that the sum of their separate modulation percentages equals 78 to 82 percent with the difference in depth of modulation of the 90 and 150 Hz signals less than 0.002 (.1 dB).

6.2.4 Standard glide slope deviation signal. The words "standard glide slope deviation signal" as used herein refer to an rf carrier amplitude modulated simultaneously by 90 Hz (±0.3%) and 150 Hz (±0.3%) signals so that the sum of their separate modulation percentages equals 78 to 82 percent with the difference in depth of modulation of the 90 and 150 Hz signals equal to 0.089 to 0.093 (1.9 to 2.1 dB).

6.2.5 Glideslope full scale deviation signal. The words "glideslope full scale deviation signal" as used herein refer to a glideslope test signal in which the difference in depth of modulation of the 90 Hz and 150 Hz is 0.175 ±0.0002 (4 ± 0.01 dB).

6.2.6 Standard glideslope deflection. The words "standard glideslope deflection" as used herein refer to the ID-1347C/ARN-82 Course Deviation Indicator glideslo~e deviation needle position produced by a glideslope deviation current of 78 microamperes. See illustration provided in 6.2.9.

6.2.7 Center circle. The words "center circle" as used herein refer to the circle surrounding the center position of the ID-1347C/ARN-82 Course Deviation Indicator. Thirty microampere of glideslope deviation current is required to deflect the glide slope deviation needle to the periphery of the center circle. See illustration provided in 6.2.9.

6.2.8 Maximum deflection. The words "maximum deflection" as used herein refer to the ID-1347C/ARN-82 Course Deviation Indicator glideslope deviation needle position produced by a deviation current of 150 microampere. See illustration provided in 6.2.9.



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6.2.9 ID-1347C/ARN-82 deflection indications. The face of the ID-1347C/ARN-82 (shown on the following page) contains the center circle and deviation dots which are used herein to describe glideslope deviation needle deflections. Beneath the illustration of the ID-1347C/ARN-82 is a table which reflects the deviation current required to deflect the glideslope deviation needle to each of the deflections indicated.

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Army - (AV)

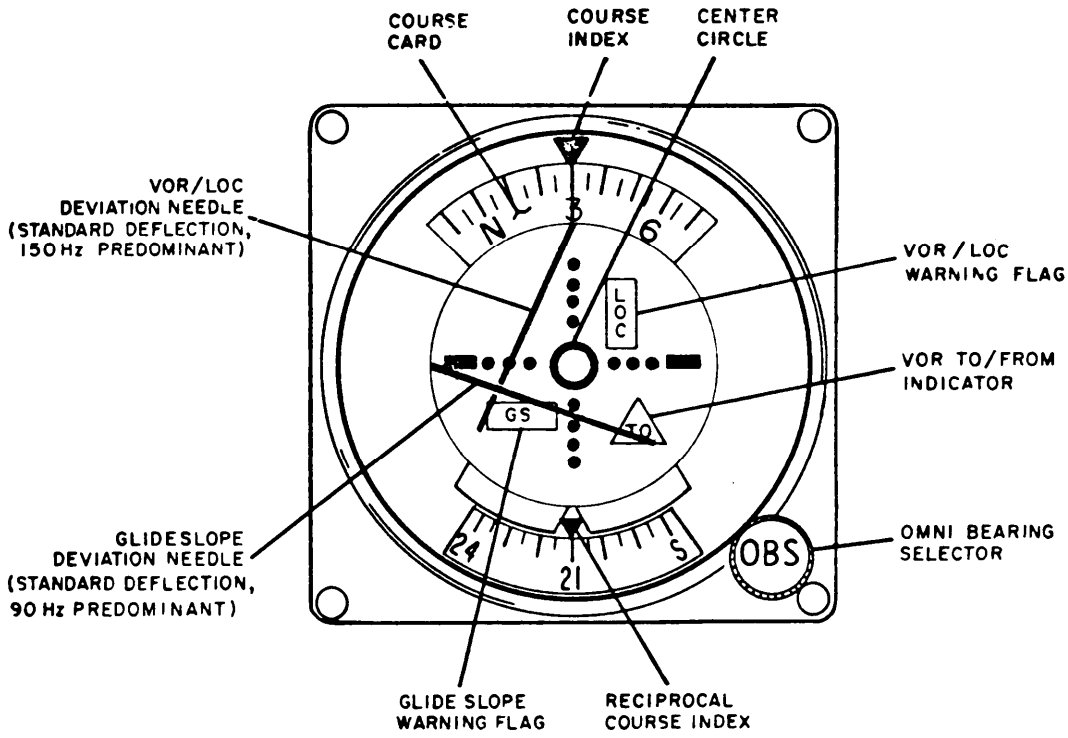
Preparing Activity

Army - (AV)

Project No. 5826-A131



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| DEFLECTION TO: | DEVIATION CURRENT REQUIRED: (in microampere) |
|---|---|
| Center of circle | 0 |
| Periphery of center circle | 30 |
| First dot | 60 |
| Second dot | 90 |
| Third dot | 120 |
| Fourth dot | 150 |

See Paragraph 6.2.9



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CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN

Receiver, Radio R-1963/ARN Serial No. _____

BENCH TESTS TEST CONDUCTED BY: _____ DATE : _____

| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.5.1 | Input power | | |
| | DC voltage | 24.75 to 30.25 volts | |
| 4.5.2 | Equipment conditions | Free of dirt, dust, etc. ; no external or internal damage. | |
| 4.5.3 | Glideslope receiver | | |
| 4.5.3.1 | Receiver sensitivity | 47 to 95 microampere deviation for 25 microvolt standard glideslope deflection signal at all specified frequencies. | |
| 4.5.3.2 | AGC characteristics | ±11 microampere deviation or less for inputs from 100 to 10,000 microvolt | |
| 4.5.3.3 | Course centering | 0 ± 15 microampere deviation | |
| 4.5.3.4 | Deflection sensitivity 90 Hz > 150 Hz | 60 to 95 microampere deviation left | |
| | 150 Hz > 90 Hz | 60 to 95 microampere deviation right | |

FIGURE l-A. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN.



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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.5.3.5 | Warning flag<br>90 Hz and 150 Hz<br>150 Hz only<br>90 Hz only | <br>$\geqslant$ 235 microampere<br>$\leqslant$ 180 microampere<br>$\leqslant$ 180 microampere | |
| 4.5.4 | Marker receiver | | |
| 4.5.4.1 | Threshold sensitivity<br>High sensitivity<br><br>Low sensitivity | <br>350 to 700 microvolt<br><br>1000 to 2100 microvolt | |
| 4.5.4.2 | Audio output power | 40 to 200 milliwatts at ten times microvolt input for 10 milliwatts. | |
| 4.5.4.3 | Lamp actuation<br>400 Hz<br><br><br>1300 Hz<br><br><br>3000 Hz | <br>Blue- on<br>Amber-off<br>White-off<br><br>Blue-off<br>Amber-on<br>White-off<br><br>Blue-off<br>Amber-off<br>White-on | |

FIGURE l-A. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



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CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN

ENGINEERING APPROVAL TESTS TEST CONDUCTED BY: _____ DATE _____

U.S. GOVERNMENT WITNESS: _____ DATE: _____

INSTALLED IN: _____ AIRCRAFT TAIL NO: _____

| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.6.1 | Preflight | | |
| 4.6.1.1 | Electrical power source | 24.75 to 30.25 volts dc | |
| 4.6.1.2 | Glide slope operation | | |
| 4.6.1.2.1 | Receiver sensitivity | Down deflection to between first and third deviation dots and flag of view for 25 microvolt input at all specified frequencies. | |
| 4.6.1.2.2 | Course centering | Deflection ≤ center circle | |
| 4.6.1.2.3 | Deflection sensitivity<br>90 Hz > 150 Hz - | Down deflection to between first and third deviation dots | |
| | 150 Hz > 90 Hz | Up deflection to between first and third deviation dots | |
| 4.6.1.2.4 | Warning signal<br>90 Hz and 150 Hz<br>150 Hz only<br>90 Hz only | <br>Flag out of view<br>Flag in view<br>Flag in view | |

FIGURE l-B. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN.



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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.6.1.3 | Marker Receiver | | |
| 4.6.1.3.1 | Threshold sensitivity | | |
| | High sensitivity | Audible tone with 700 microvolt input | |
| | Low sensitivity | Audible tone with 2100 microvolt input | |
| 4.6.1.3.2 | Audio output power | Audible tone which is variable and intelligible | |
| 4.6.1.3.3 | Lamp actuation | THREE LAMPS / SINGLE LAMP | |
| | 400 Hz | Blue-on<br>Amber-off On<br>White-off | |
| | 1300 Hz | Blue-off<br>Amber-on On<br>White-off | |
| | 3000 Hz | Blue-off<br>Amber-off On<br>White-on | |

FIGURE l-B. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:47Z
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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.6. 1.5 | Glideslope system operation | | |
| | Centering | Deflection ≤ center circle | |
| | Deflection | | |
| | 90Hz > 150 Hz | Deflection from center then down to between third and fourth deviation dots. | |
| | 150 Hz > 90 Hz | Deflection from center then up to between third and fourth deviation dots. | |
| | No 150 Hz | Glideslope warning flag in view | |
| 4.6.1.6 | Marker beacon operation | THREE LAMPS / SINGLE LAMP | |
| | 400 Hz | Blue-on<br>Amber-off<br>White-off | on |
| | 1300 Hz | Blue-off<br>Amber-on<br>White-off | on |
| | 3000 Hz | Blue-off<br>Amber-off<br>White-on | on |
| | Audio | Intelligible | |

FIGURE l-B. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.6.2 | Flight | | |
| 4.6.2.1 | Glideslope operation | | |
| | a. Glideslope path encountered | Glideslope warning flag out of view | |
| | b. Off center path below | Glide slope deviation needle deflected up | |
| | above | Glide slope deviation needle deflected down | |
| 4.6.2.1.2 | Glideslope antenna field pattern and system sensitivity 15 nautical miles from station, 30° right and left angles to localizer center line | Glideslope warning flag out of view, glide-slope deviation needle deflects up and down with respect to glide-slope path. | |
| 4.6.2.2 | Marker beacon operation | | |
| 4.6.2.2.1 | Receiver, Radio R-1963 ARN marker beacon flight requirements | Lamp(s) light and audio present as specified | |
| 4.6.2.2.2 | Marker beacon antenna pattern and system sensitivity flight requirements | Lamp(s) light and audio present as specified | |

FIGURE l-B. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.6.2.3 | ILS flight check | | |
| 4.6.2.3.1 | Marker beacon performance | Lamp(s) light and audio present as specified | |
| 4.6.2.3.2 | Glideslope performance | | |
| | Over outer and middle markers | Proper altitude indicated | |
| | From 700 feet to decision altitude | Track center of glide-slope path to within first dot deflection or ±22 feet whichever is the larger | |
| | Flying centered glideslope needle to runway | Aircraft lands on glide-slope reference datum of runway | |

FIGURE l-B. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



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CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN

PRODUCTION APPROVAL TESTS TEST CONDUCTED BY: _____ DATE: _____

INSTALLED IN _____ AIRCRAFT TAIL NO. _____

| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.7.1 | Preflight | | |
| 4.7.1.3 | Glideslope operation<br>Centering | Deflection center circle | |
| | Deflection<br>90 Hz > 150 Hz | Deflection from center to down between third and fourth deviation dots with no hang-ups | |
| | 150 Hz > 90 Hz | Deflection from center to up between third and fourth deviation dots with no hang-ups | |
| | No 150 Hz | Glideslope warning flag in view | |
| 4.7.1.4 | Marker beacon operation | THREE LAMPS / SINGLE LAMP | |
| | 400 Hz | Blue-on<br>Amber-off On<br>White-off | |
| | 1300 Hz | Blue-off<br>Amber-on On<br>White-off | |
| | 3000 Hz | Blue-on<br>Amber-off On<br>White-on | |
| | Audio | Intelligible | |
| | Test energized | All marker lamps on | |

FIGURE l-C. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN.



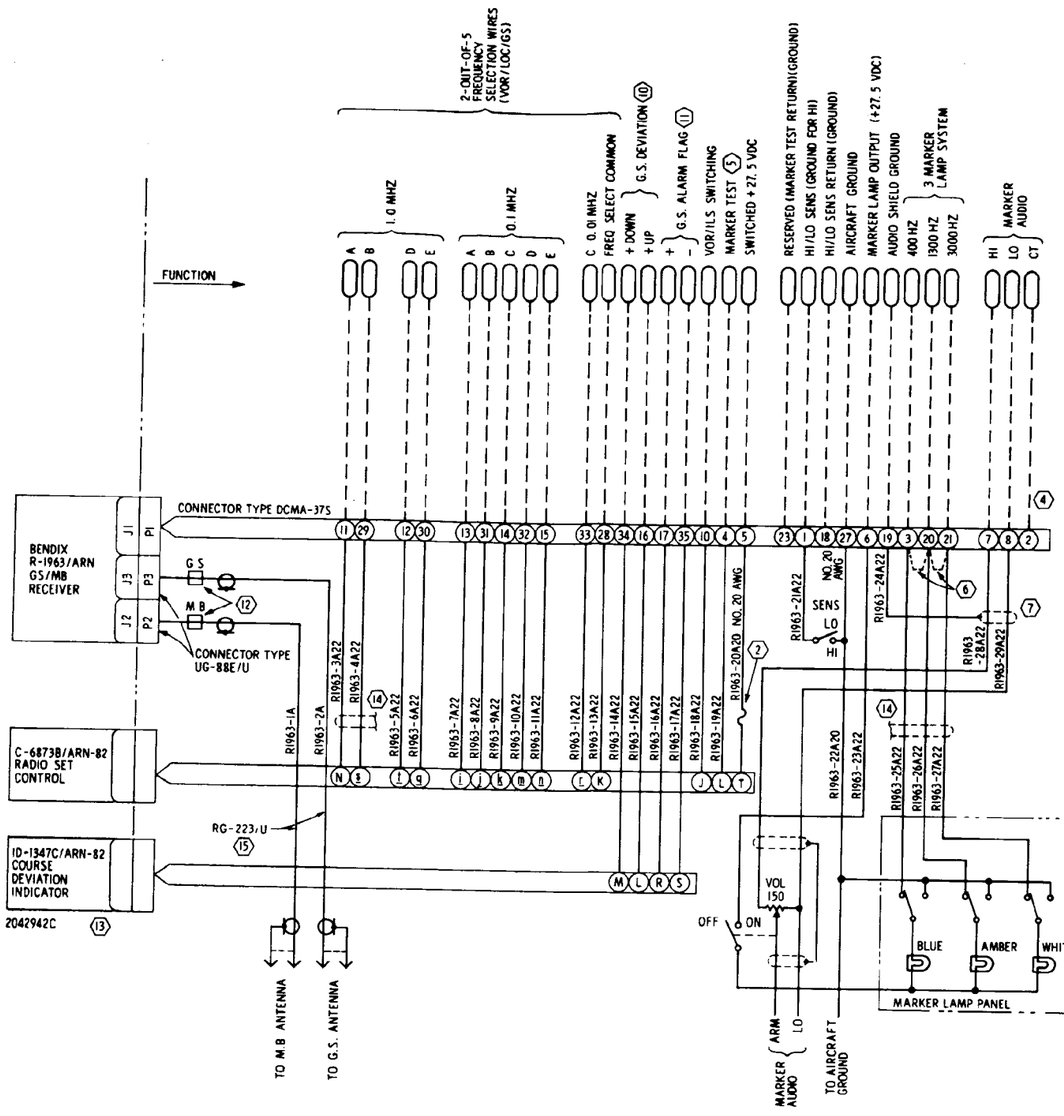
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| APPLICABLE PARAGRAPH | TEST | MINIMUM STANDARDS | REMARKS |
|---|---|---|---|
| 4.7.2 | Flight | | |
| 4.7.2.1 | Glideslope operation | | |
| | a. 15 nautical miles from station | Glideslope warning flag out of view | |
| | b. Off center path Below | Glideslope deviation needle deflected up | |
| | Above | Glideslope deviation needle deflected down | |
| 4.7.2.4 | Production flight test Marker beacon operation | Lamp(s) light and audio present as specified | |
| 4.7.2.5 | ILS flight check | | |
| | a. Flight path encountered | VOR/LOC and glideslope warning flags out of view | |
| | b. Over marker beacons | Correct marker beacon and altitude indications as specified | |
| | c. From 700 feet to decision altitude | Track center of glide-slope path to within first deviation dot or ±22 feet, whichever is larger. | |
| | d. Flying centered glideslope and VOR/LOC deviation needles | Bring aircraft to localizer and glideslope reference datums on runway. | |

FIGURE 1-C. CHECK-OFF LIST FOR TEST OF RECEIVER, RADIO R-1963/ARN. (Continued)



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NOTES:

1. ALL WIRES NO. 22 AWG STRANDED UNLESS OTHERWISE SPECIFIED.
2. USE 1.0 AMPERE CIRCUIT BREAKER.
3. PINS 9, 22, 24, 25, 26, 36, 37 ARE SPARES IN CONNECTOR J1.
4. CENTER TAP (C.T.) ON J1 PIN 2 IS AVAILABLE, BUT NOT CONNECTED.
5. CONTROL C-6873B/ARN-82 SUPPLIES GROUND WHEN TEST SWITCH IS ENERGIZED.
6. MARKER LAMP CONNECTIONS SHOWN ARE FOR A THREE MARKER LAMP SYSTEM. FOR A SINGLE MARKER LAMP SYSTEM, CONNECT DASHED LINES AND USE A SINGLE GREEN MARKER LAMP. MARKER LAMPS SHALL BE TYPE MS-25041 (WITH DIMMER FEATURE) OR EQUIVALENT.
7. ALL AUDIO SHIELDS SHALL BE TERMINATED AT P1 END, PIN 19, ONLY.
8. INSTALLATION SHALL BE ACCORDANCE WITH APPLICABLE ISSUE OF SPECIFICATION MIL-W-5088.
9. WIRE SIZES SHOWN ARE MINIMUM SIZES ONLY AND LARGER SIZES SHALL BE USED AS REQUIRED.
10. THE GLIDE SLOPE DEVIATION INDICATOR OUTPUT CIRCUIT IS CAPABLE OF DRIVING UP TO THREE 1000 OHM, 150-0-150 MICROAMPERE LOADS. NO SUBSTITUTE LOADS ARE REQUIRED IF LESS THAN THREE LOADS ARE USED.
11. THE GLIDE SLOPE WARNING FLAG OUTPUT CIRCUIT IS CAPABLE OF ACTUATING THREE 1000 OHM 250 MICROAMPERE LOADS. NO SUBSTITUTE LOADS ARE REQUIRED IF LESS THAN THREE LOADS ARE REQUIRED.
12. TO PREVENT CROSS-CONNECTIONS, ATTACH A NON-METALLIC FLAG (OR EQUIVALENT) TO COAX CABLE TO IDENTIFY COAX CONNECTORS TO R-1963/ARN RECEIVER.
13. ID-1347C/ARN-82 SWITCH POSITIONED AS REQUIRED BY VOR/LOC RECEIVER.
14. SHIELDING REQUIREMENTS AND/OR EMC CONTROL METHODS SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF MIL-E-6051. GROSS RFI SHIELDED CABLES SHALL BE UTILIZED FOR ALL INTERCONNECTING CABLES. THE CABLE SHALL HAVE A SHIELD WEAVE PROVIDING NO LESS THAN 95% COVERAGE AND A SHIELDING EFFECTIVENESS OF APPROXIMATELY 70 DB FROM 50 TO 100 MHZ USING METEX 05-08600342-01, ONE-INCH WIDE SHIELD TAPE (FROM THE METEX CORPORATION, EDISON, NEW JERSEY) OVERLAPPED 50%, OR EQUIVALENT. ALL SHIELDED CABLES SHALL HAVE THE SHIELD TERMINATED AT EACH END IN THE SPECIFIED CONNECTORS WITH AN EMI BACKSHELL TO PROVIDE PERIPHERAL BONDING OF THE SHIELD. RECOMMENDED EMI BACKSHELLS (FROM GLENAIR, INC., GLENDALE, CALIFORNIA ARE:

| CONNECTOR | GLENAIR PART NO. (OR EQUIVALENT) |
|---|---|
| P1 | G8209-4Y2 |

THE EMI BACKSHELL SHALL BE SUPPLIED WITH MOUNTING SCREWS (CAPTIVATED), STRAIN RELIEF, CABLE FERRULE, AND METAL IMPREGNATED GASKET. INSTALLATION WITHOUT GROSS RFI SHIELDED CABLES AND EMI BACKSHELLS IS PERMISSIBLE PROVIDED APPROPRIATE EMC TESTING IS PERFORMED TO INSURE EMI COMPATIBILITY WITH OTHER AIRCRAFT SYSTEMS. ALL EMC REQUIREMENTS OF THE SPECIFICATION MIL-I-49076(EL) SHALL BE MET. FINAL APPROVAL SHALL BE GRANTED BY THE GOVERNMENT.

15. INSTALLATION WITHOUT DOUBLE SHIELDED RF CABLES (RG-223/U OR EQUIVALENT), USING RG-58C/U OR EQUIVALENT AS REPLACEMENT, IS PERMISSIBLE PROVIDED APPROPRIATE EMC TESTING IS PERFORMED TO INSURE EMI COMPATIBILITY WITH OTHER AIRCRAFT SYSTEMS. ALL EMC REQUIREMENTS OF THE SPECIFICATION MIL-I-49076(EL) SHALL BE MET. FINAL APPROVAL SHALL BE GRANTED BY THE GOVERNMENT.

FIGURE 2. RECEIVER, RADIO R-1963/ARN, INTERWIRING DIAGRAM



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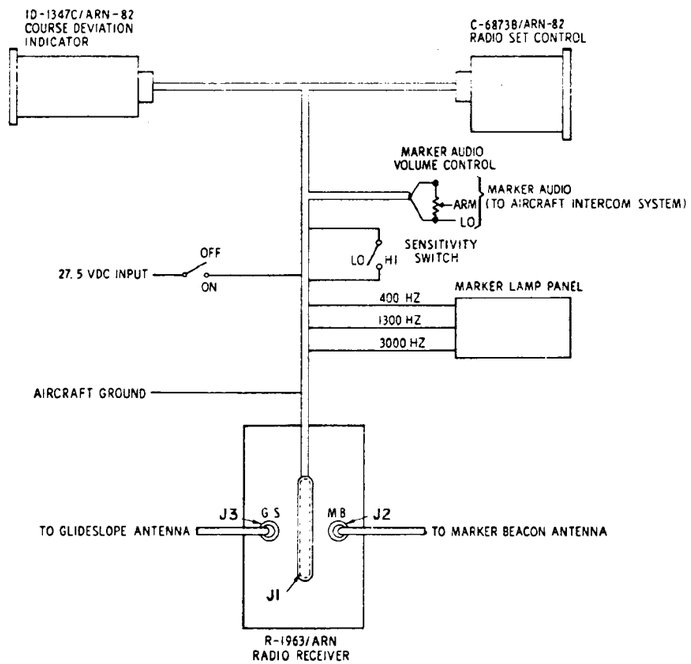
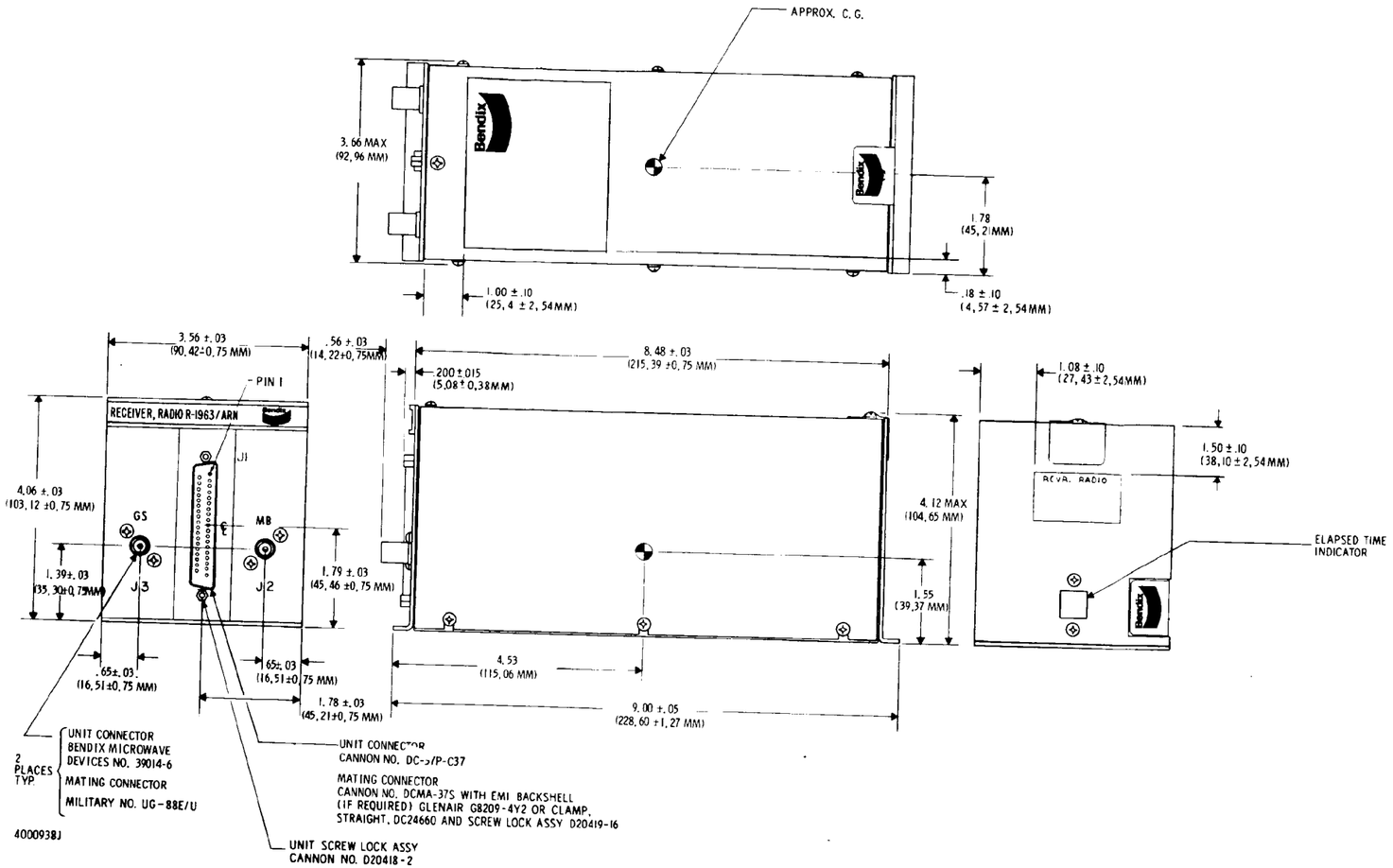



FIGURE 3. RECEIVER, RADIO R-1963/ARN, CABLING DIAGRAM



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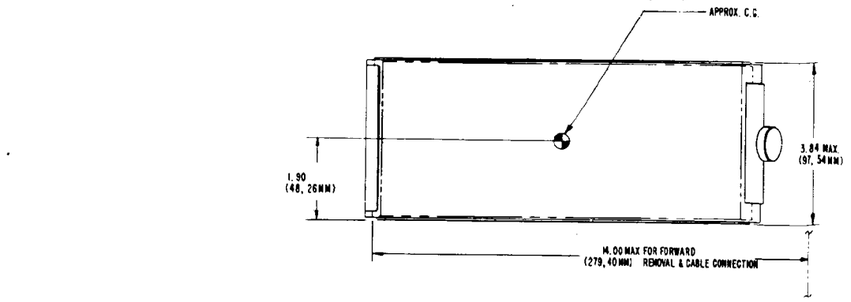

NOTES:

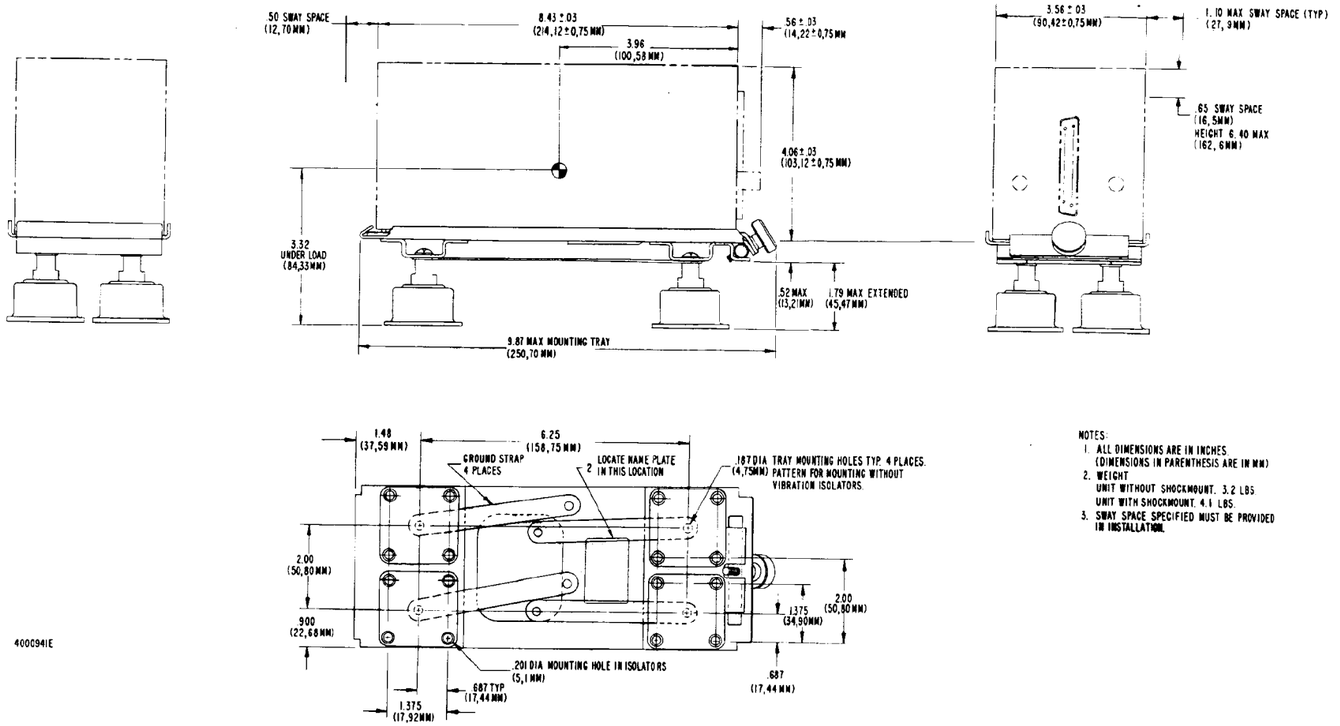
1. HEAT DISSIPATION: 7.5 WATTS MAX.
2. ALL DIMENSIONS ARE IN INCHES (DIMENSIONS IN PARENTHESIS ARE IN MM)
3. WEIGHT: 3.2 LBS. MAX.
4. STAMP WARRANTY TERMINATION DATE AND RECEIVER SERIAL NO. ON WARRANTY LABEL.
5. STAMP WEIGHT AND SERIAL NO. ON NAMEPLATE PRIOR TO SHIPPING.

FIGURE 4. RECEIVER, RADIO R-1963/ARN. OUTLINE DRAWING



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NOTES:
1. ALL DIMENSIONS ARE IN INCHES
(DIMENSIONS IN PARENTHESIS ARE IN MM)
2. WEIGHT
UNIT WITHOUT SHOCKMOUNT, 3.2 LBS
UNIT WITH SHOCKMOUNT, 4.1 LBS
3. SWAY SPACE SPECIFIED MUST BE PROVIDED
IN INSTALLATION.

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FIGURE 5. RECEIVER, RADIO R-1963/ARN WITH MOUNTING BASE, ELECTRICAL EQUIPMENT MT-4835/ARN, OUTLINE DRAWING



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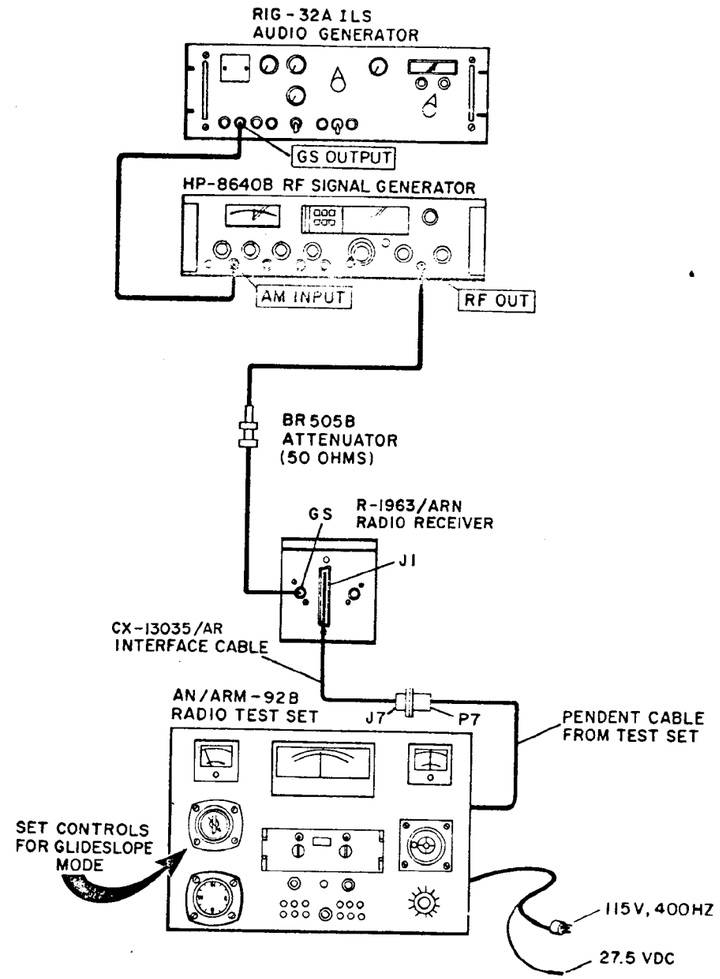



FIGURE 6. RECEIVER, RADIO R-1963/ARN GLIDESLOPE RECEIVER BENCH TEST SETUP



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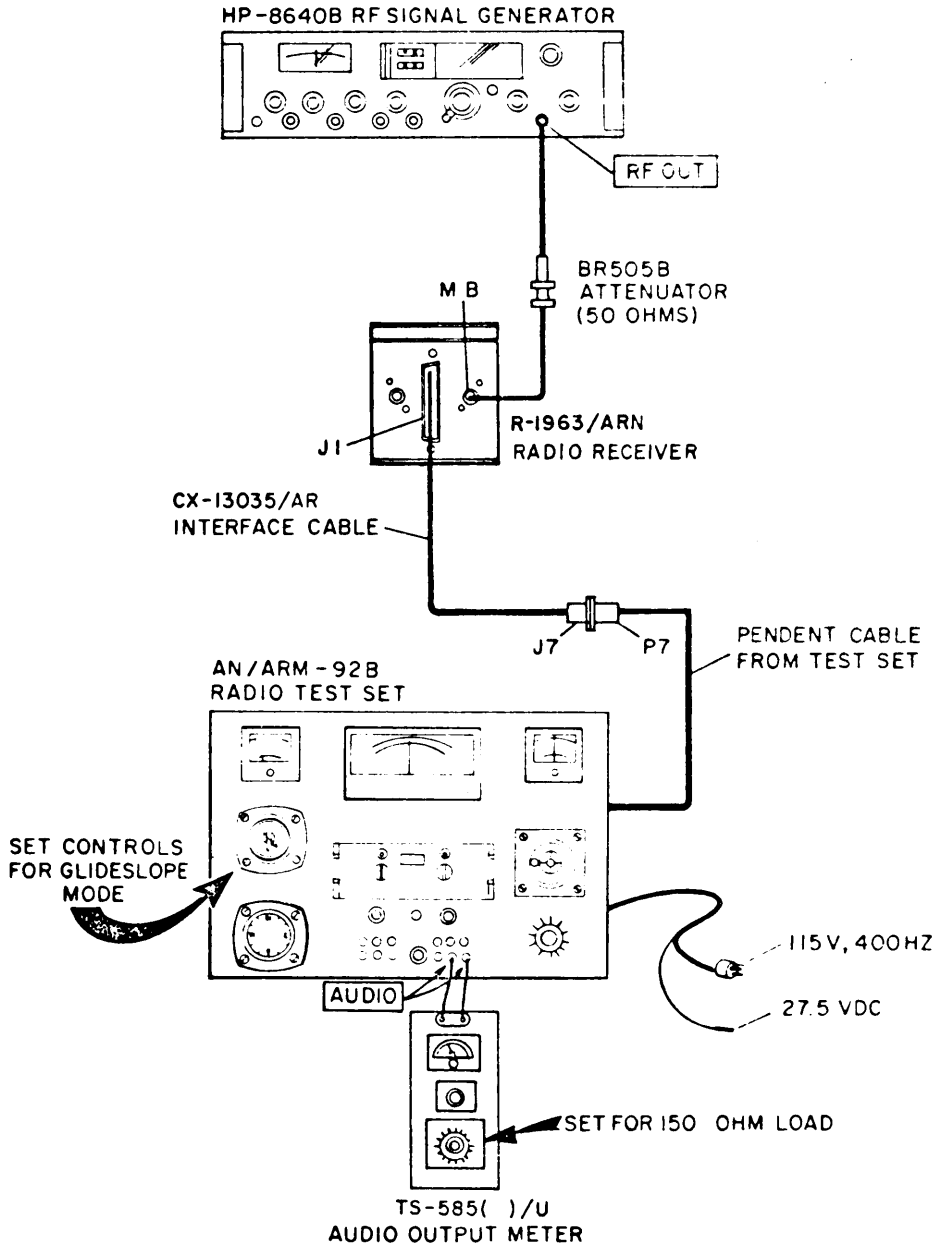



FIGURE 7. RECEIVER, RADIO R-1963/ARN
MARKER BEACON RECEIVER BENCH SETUP



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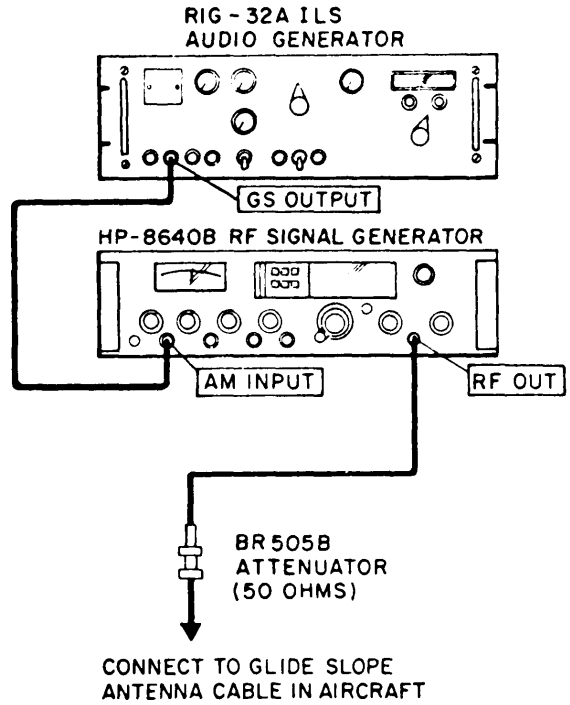



FIGURE 8. RECEIVER, RADIO R-1963/ARN, GLIDESLOPE RECEIVER TEST SETUP FOR ENGINEERING APPROVAL TESTS



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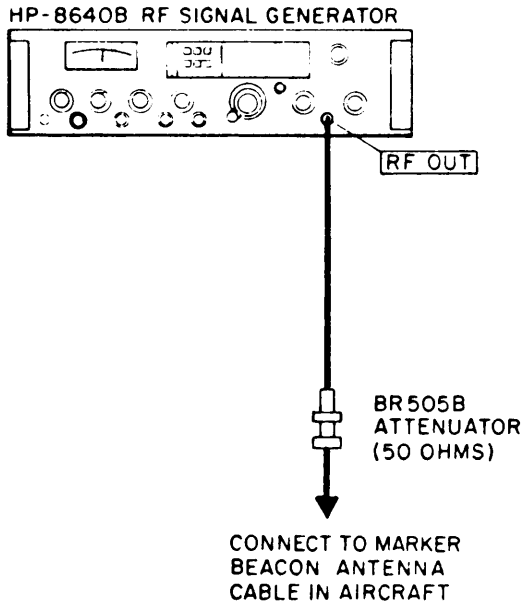
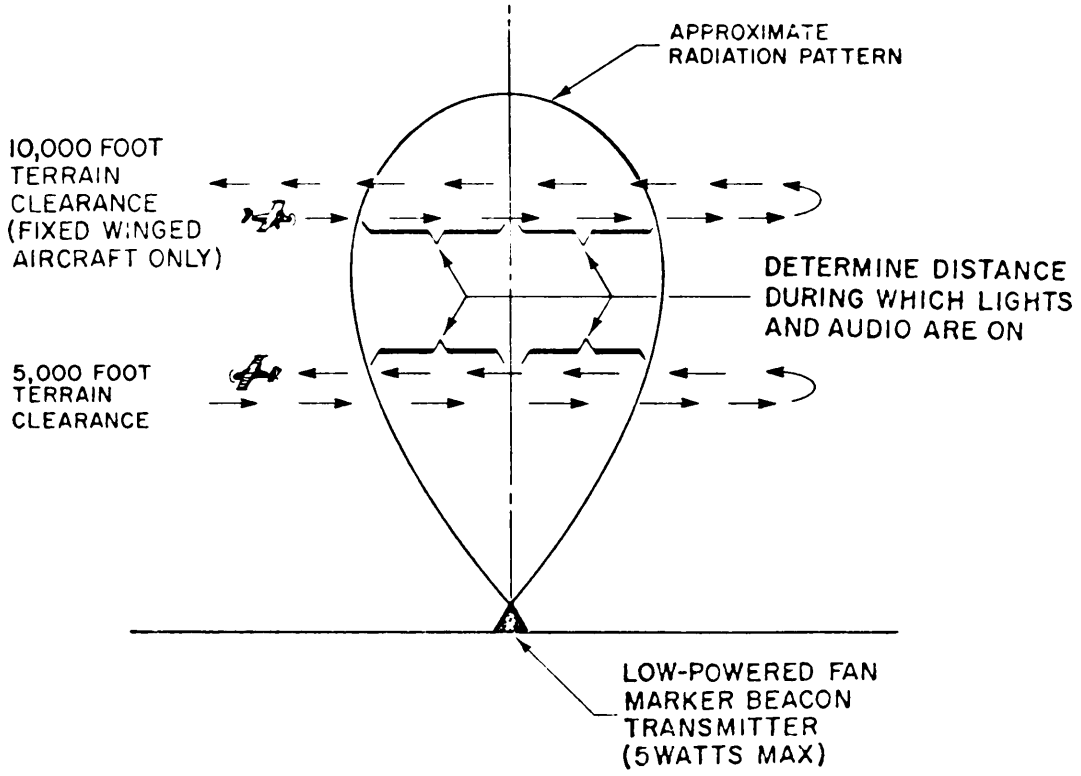



FIGURE 90 RECEIVER, RADIO R-1963/ARN, MARKER RECEIVER TEST SETUP FOR ENGINEERING APPROVAL TESTS



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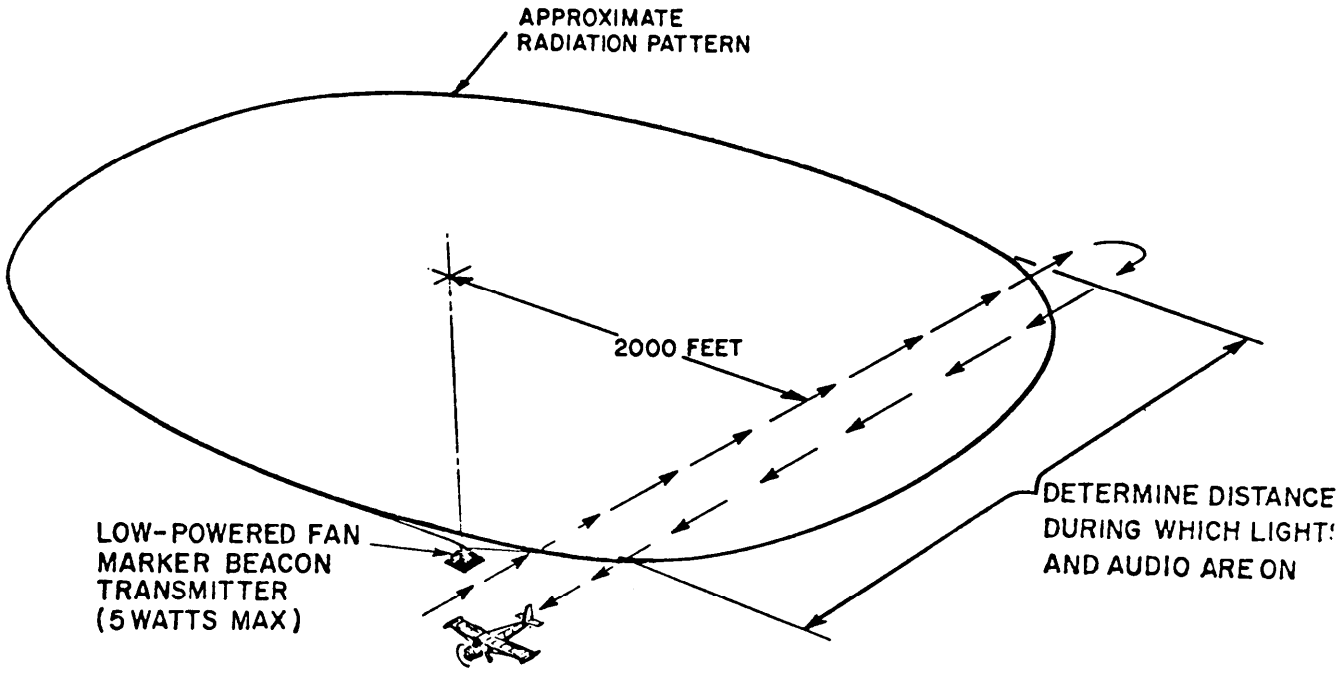



Flight Test

1. Four flight passes shall be required to complete the test pattern.
2. All passes shall be directly over the marker beacon and on the airway course.
3. Make 5000 foot passes with normal stores and landing gear extended.
4. Marker beacon receiver in HI sensitivity mode.



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Flight Test

1. Two flight passes through the transmitter pattern shall be required, one in each direction. Make passes with normal stores and landing gear extended.

2. The flight passes shall be made at 2000 feet altitude (terrain clearance) and at 2000 feet horizontal range from the transmitter. and parallel to the airway course.

3. Marker beacon receiver in HI sensitivity mode.

FIGURE 11. MARKER BEACON RADIATION PATTERN (TOP VIEW)

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